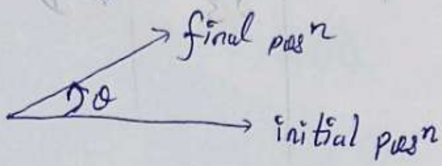
Lec - 1

Mathematical Tool

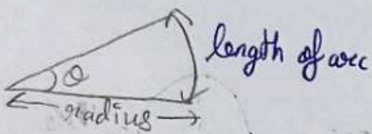
Trigonometry

1° (degree) $= 60'$ (minutes)
$1'$ (minute) $= 60''$ (seconds)
1° (degree) $= 3600''$ (seconds)

- Angle 'θ' :- The amount of rotation of a line from its initial position.
  ~~Measures~~ definitions in term of θ.



Planar angle



$$\theta = \frac{\text{length of arc}}{\text{radius}}$$

S.I unit - radian
Practical unit - degree

$1^c \rightarrow$ one radian

when length of arc is same as radius. $\theta = 1^c$, $l = r$

$$\theta = \frac{l}{r}$$

$$\theta = \frac{r}{r} = 1$$

degree $\longleftrightarrow$ Radian

$$\pi^c \longrightarrow 180^\circ$$

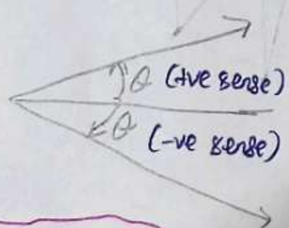
$$1^c = \frac{180^\circ}{\pi} \qquad 1^\circ = \frac{\pi}{180}$$

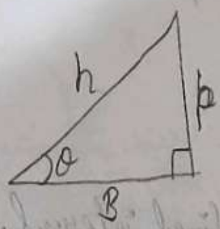
$l = \theta^c r$
$l = 2\pi r$
$2\pi r = \theta^c r$
$\Rightarrow \pi = 180^\circ$

$60^\circ = \frac{\pi}{3}^c$	$= \frac{\pi}{3}$ rad
$90^\circ = \frac{\pi}{2}^c$	$= \frac{\pi}{2}$ rad

$$\theta^c = \left(\frac{180}{\pi} \times \theta\right)^\circ$$

$$\theta^\circ = \left(\frac{\pi}{180} \times \theta\right)^c$$





$$h = \sqrt{p^2 + B^2}$$

$$\frac{P\ B\ P}{H\ H\ B}$$

- $\sin\theta \rightarrow P/H$
- $\cos\theta \rightarrow B/H$
- $\tan\theta \rightarrow P/B$
- $\text{cosec}\,\theta \rightarrow H/P$
- $\sec\theta \rightarrow H/B$
- $\cot\theta \rightarrow B/P$

$\theta = 0^\circ$

H, B, P

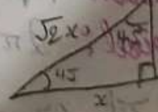
$\sin 0^\circ = P/H = 0$

~~$\cos\theta$~~

$\cos 0^\circ = B/H = 1$

$\tan 0^\circ = P/B = 0$

$\theta = 45^\circ$



$\sin 45 = \frac{1}{\sqrt{2}}$

$\cos 45 = \frac{1}{\sqrt{2}}$

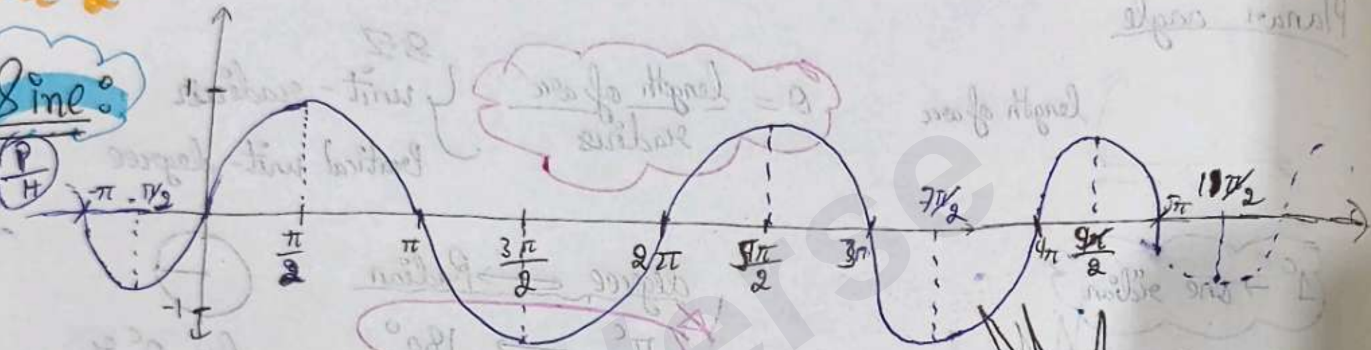
$\tan 45 = 1$

Ratio \ Angle	0°	30°	45°	60°	90°
Sin	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
Cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
Tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	N.D

Le-2

Sine:

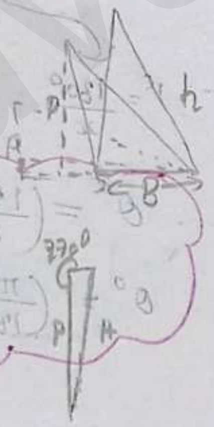
$\frac{P}{H}$



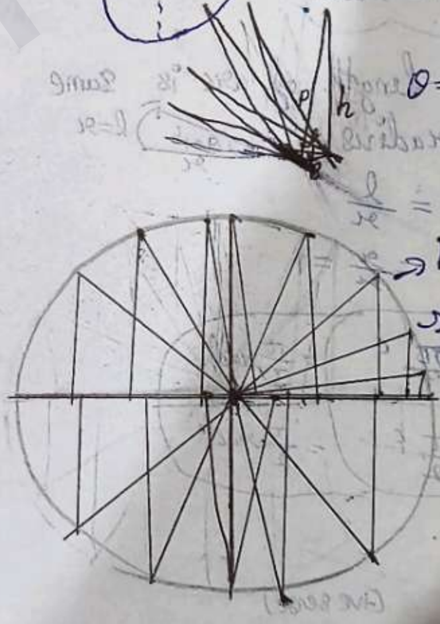
Per(↓)(↓)

$90^\circ \to 180^\circ$ } P decreases

$\sin\theta$ = value decrease



$\theta = 90^\circ$ [$h \approx P$]
[$B \approx 0$]



$180^\circ \to 270^\circ$ (P(↑))

$\sin\theta$ → − (value increase)

$270 \to 360^\circ$ (P(↓))

$\sin\theta$ → − (value decrease)

Properties of sine:

① Period function: Repeat after a fixed interval.

② Time period: 2π interval

③ $+1 \geq \sin\theta \geq -1$ or $-1 \leq \sin\theta \leq +1$

④ Sine θ may be increasing or decreasing depending upon the interval

$(0-90^\circ)$ (↑)

$(90^\circ-180^\circ)$ (↓) & $(180^\circ-270^\circ)$ (↓)

$(270^\circ-360^\circ)$ (↑)

(5) Sine is an odd function.

$\sin 30^\circ = \frac{1}{2}$

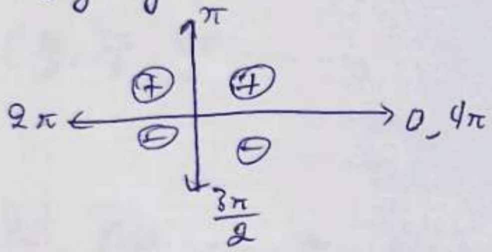
$\sin(-30^\circ) = -\frac{1}{2}$

(6) $\sin(n\pi) = 0$; n is integer

~~(7) $\sin(2n + \frac{\pi}{2}) =$~~

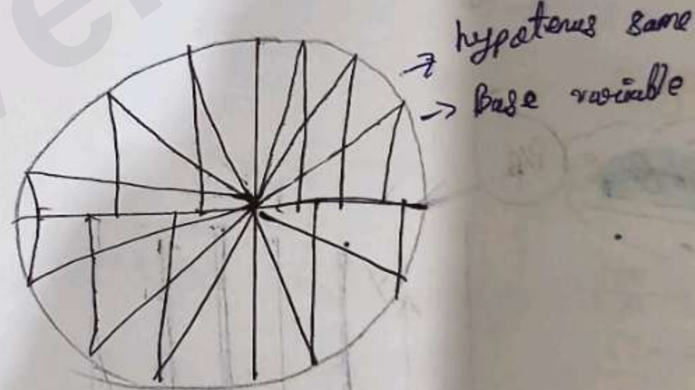
(7) $\sin(2n+1)\frac{\pi}{2} = \pm 1$

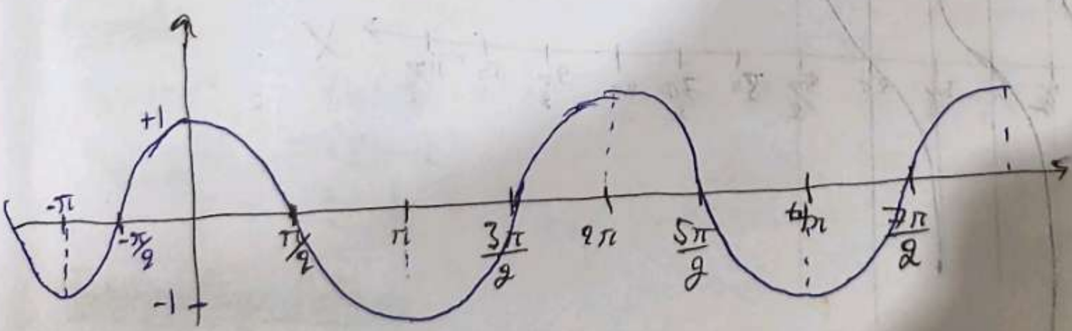
(8) Sign of sin



Cosine :- $\frac{B}{H}$

0-90°	(↓) {1 → 0}
90°-180°	{0 → -1}
180°-270°	{-1 → 0}
270°-360°	{0 → 1}





Property of Cosine :

(1) Periodic function : Repeat after a fixed interval

(2) Time period : 2π interval

(3) $-1 \leq \cos\theta \leq 1$

④ Cos θ may be increasing or decreasing depending upon the interval.

(0-90°)↑
(90-180°)↓ (180-270) ↑
(270-360°)↑

⑤ Cos θ is an even function

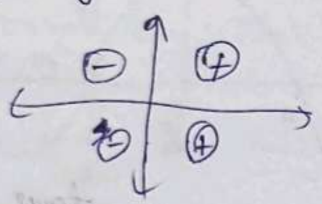
$\cos 30° = \frac{1}{2}$

$\cos(-30°) = \frac{1}{2}$

⑥ $\cos(n\pi) = \pm 1$

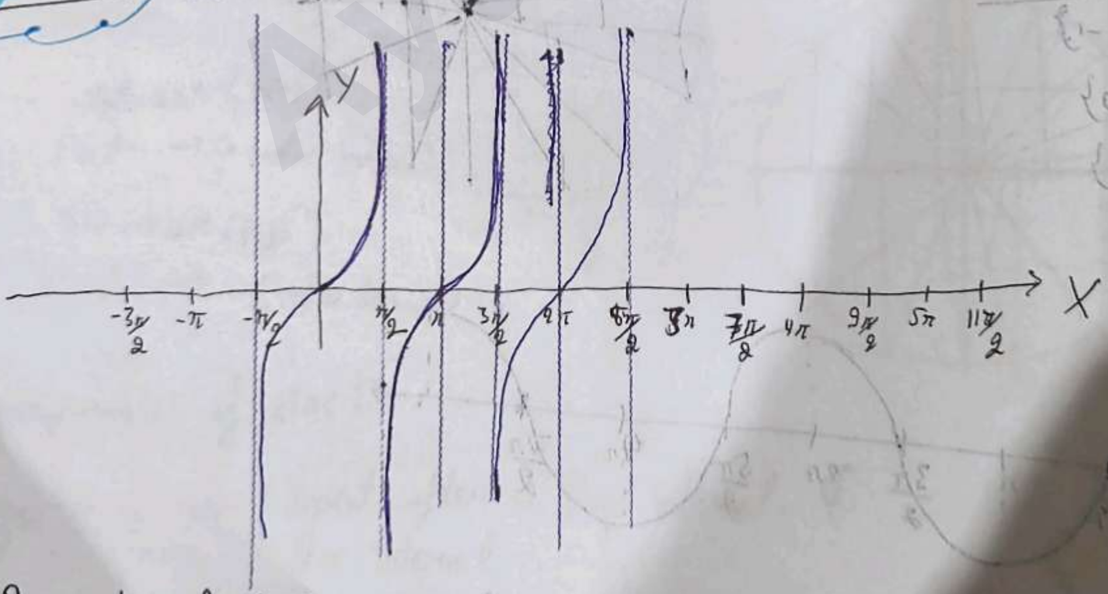
⑦ $\cos (2n+1)\frac{\pi}{2} = 0$

⑧ Sign of Cos



Lec-3

Tan θ : P/B



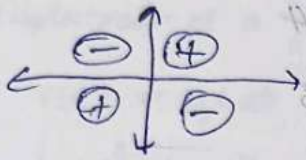
Property of Tan:

① Period function
② Time interval : π
③ $-\infty < \tan\theta < \infty$
④ Odd function: $\tan(-\theta) = -\tan\theta$

5. $\tan(n\pi) = 0$

6. $\tan(2n \pm 1)\frac{\pi}{2}$ = not defined

7. Sign of Tan :



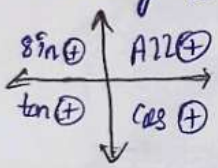
How to find T-ratio's for bigger angle

1. $(\pi, 2\pi, 3\pi, 4\pi, 5\pi, \ldots\ldots) \longrightarrow$ No change in argument

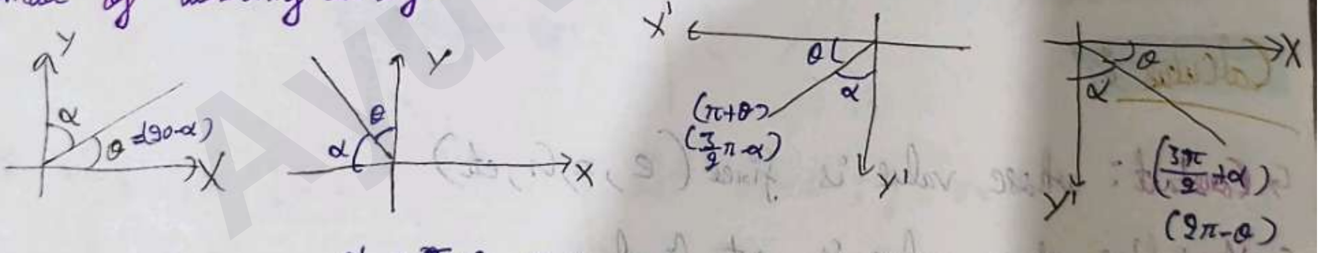
2. $\left(\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots\ldots\right) \longrightarrow$ Argument changes

$$\left\{ \begin{array}{l} \sin \longleftrightarrow \cos \\ \tan \longleftrightarrow \cot \\ \sec \longleftrightarrow \text{cosec} \end{array} \right\}$$

2. Put sign according to ASTC rule



Formate of writing an angle

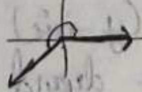


$\alpha = \frac{\pi}{2} + \theta = \pi - \alpha$

Examples

$\sin 135^\circ = \sin(180 - 45^\circ)$
$= +\sin 45^\circ$
$= \frac{1}{\sqrt{2}}$

$\sin 225^\circ = \sin(~~360~~\ 180 + 45)$
$= -\cos 45$
$= -\frac{1}{\sqrt{2}}$

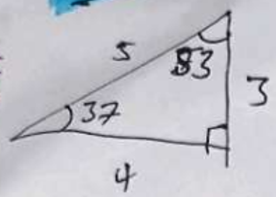


Trigonometry Ratio

1. $\sin^2\theta + \cos^2\theta = 1$
2. $1 + \tan^2\theta = \sec^2\theta$
3. $1 + \cot^2\theta = \text{cosec}^2\theta$
4. $\sin 2\theta = 2\sin\theta\cos\theta$
5. $\sin(A+B) = \sin A \cos B + \cos A \sin B$
6. $\sin(A-B) = \sin A \cos B - \cos A \sin B$
7. $\cos(A+B) = \cos A \cos B - \sin A \sin B$
8. $\cos(A-B) = \cos A \cos B + \sin A \sin B$
9. $\cos 2\theta = \cos^2\theta - \sin^2\theta = (\cos\theta + \sin\theta)(\cos\theta - \sin\theta)$
10. $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
11. $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
12. $\tan(2A) = \frac{2\tan A}{1 - \tan^2 A}$

Lec-4

Cute Triangle



$\sin 37 = 3/5$ $\qquad$ $\tan 37 = 3/4$

$\cos 53 = 3/5$ $\qquad$ $\tan 53 = 4/3$

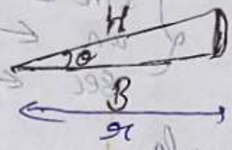
$\sin 53 = 4/5$

$\cos 53^\circ = 3/5$

Small Angle Approximation:

$\theta < 10^\circ$

① $\sin\theta = \frac{P}{H} = \frac{r\theta}{r} \approx \theta$



$P \approx$ length of arc $\approx r\theta$

② $\cos\theta \approx 1$

③ $\tan\theta \approx \theta$

④ $\sin\theta \approx \tan\theta$

→ θ in radian

Calculus:

↳ Constant: whose value is fixed (e, π, G, etc)

↳ Variable: whose value is not fixed.

Independent variable (x-axis)

Dependent Variable (y-axis)
value depend on independent variable

Relation:

A relation defines the order dependency of D.V on I.V.

- $y \propto x$ → linear relation
- $y \propto x^2$ → Quadratic relation
- $y \propto x^3$ → Cubic relation
- $y \propto \frac{1}{x}$ → Inverse relation
- $y \propto \frac{1}{x^2}$ → Inverse square relation

- $y \propto e^x$ → exponential relation
- $y \propto \log x$ → logarithmic relation
- $y \propto \sin x$ → trigonometric relation

Function:

A mathematical expression which defines an exact relationship b/w D.V an...

Ex - $s(t) = ut + \frac{1}{2}at^2$, u, a = constant

↓ displacement as a function of time

$v(t) = u + at$

↳ velocity as a function of time

Lec-5

Linear Relationship

$y \alpha x$ → always pass through origin
→ straight line

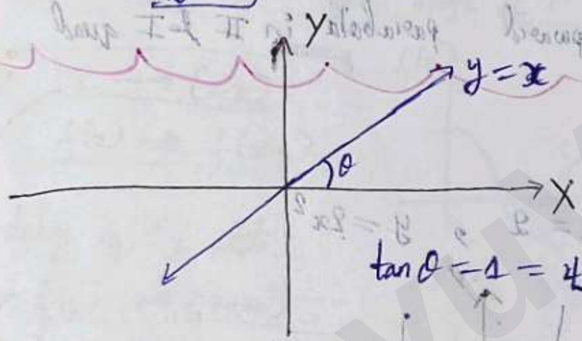
$y = mx$
↳ proportionality const.

$\boxed{-\infty < m \leq 0}$ → slope in II & IV quad.

$\boxed{\infty > m \geq 0}$ → slope in I & III quad.

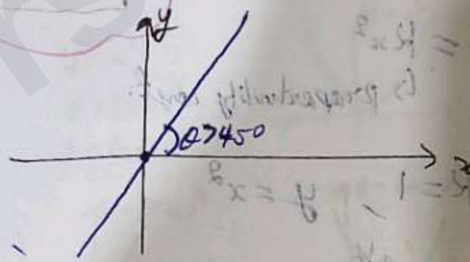
Case I

Let $m = 1$; $\boxed{y = x}$

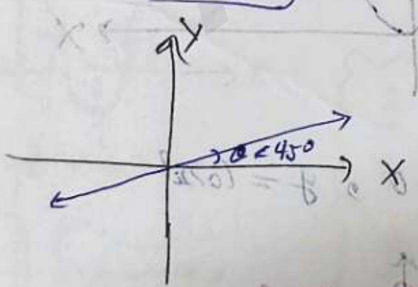


$\tan\theta = 1 = 45°$

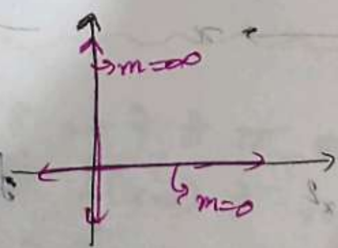
Let $m = 2$; $\boxed{y = 2x}$

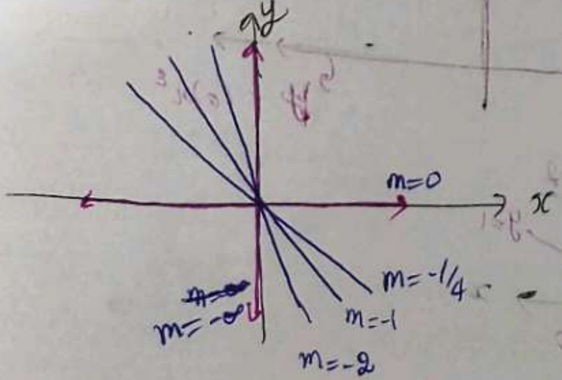


Let $m = \frac{1}{2}$; $\boxed{y = \frac{1}{2}x}$



Let $m = 0$; $y = 0x$

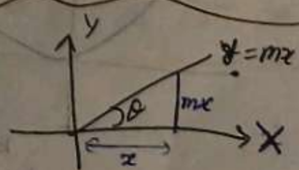




$\tan\theta = \frac{mx}{x}$

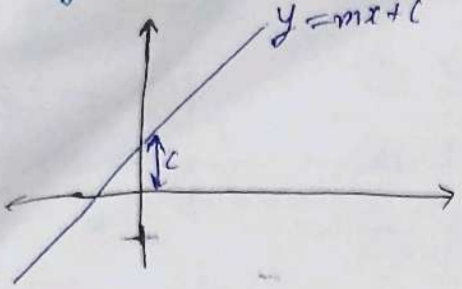
$\boxed{\tan\theta = m}$

$\boxed{\theta = \tan^{-1}(m)}$

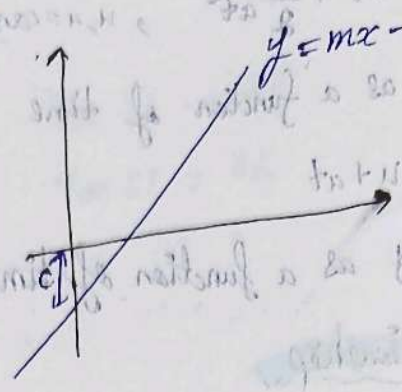


Addition of constant:

$y = mx + c$



$y = mx - c$



~~Straight~~ ($y=mx+c$), Straight line having intercept at "c"

Quadratic Relationship

$y \propto x^2$

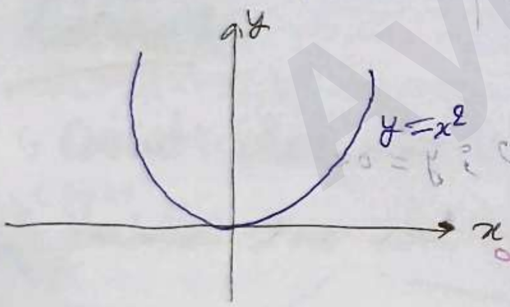
$\boxed{-\infty < k \leq 0} \rightarrow$ Downward parabola in III & IV quad.

$\boxed{0 \leq k < \infty} \rightarrow$ Upward parabola in II & I quad

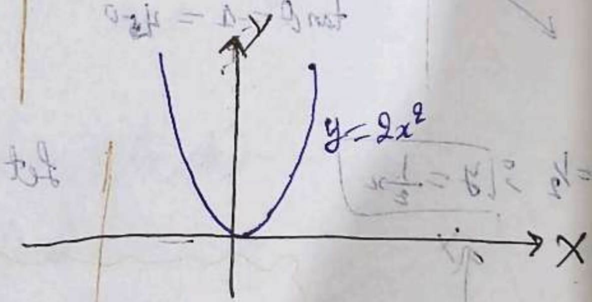
$y = kx^2$

↳ proportionality const.

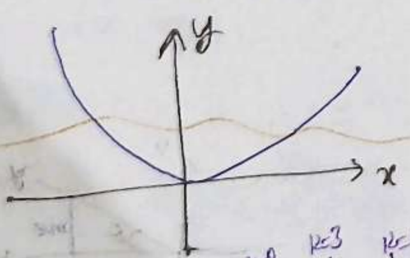
Let $k=1$, $y = x^2$



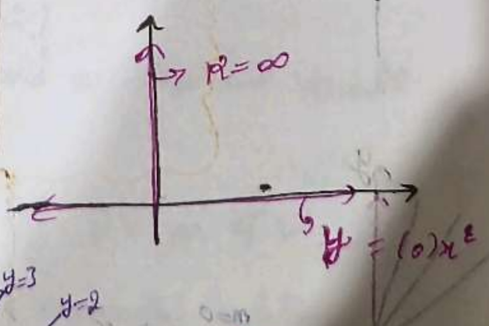
Let $k=2$, $y=2x^2$

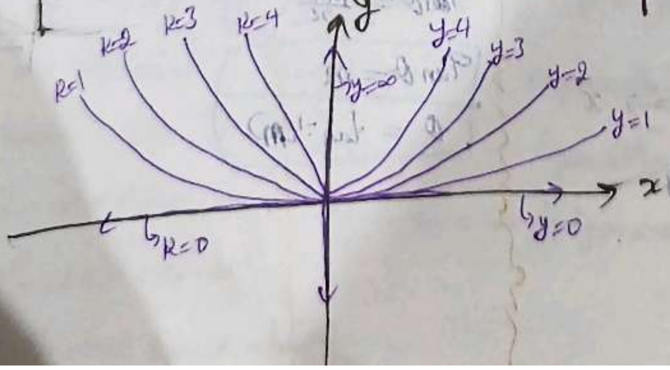


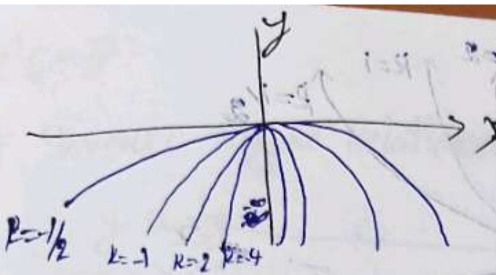
Let $k=1/2$, $y=\frac{1}{2}x^2$



Let $k=0$, $y=(0)x^2$





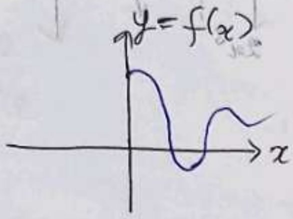


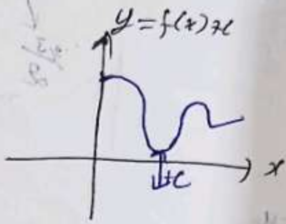
Lec-6

Addition of constant :

$$y = f(x) + c$$

$$y = f(x)$$





if $c = +ve$

whole curve will get shift of 'c' units upward.

if $c = -ve$

whole curve will get shift of 'c' units downward

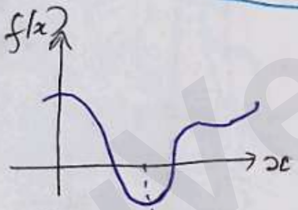
note: $\boxed{x \rightarrow (x-a)}$

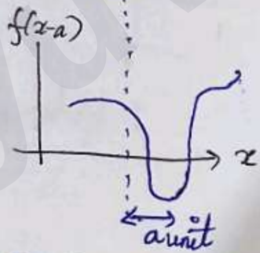
$f(x) \rightarrow f(x-a)$

delay by "a" unit

$f(x) \rightarrow f(x+a)$

advance by "a" unit

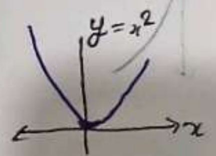


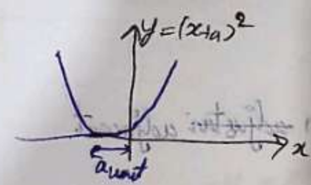


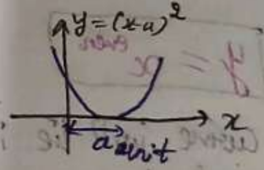
$y = x^2$

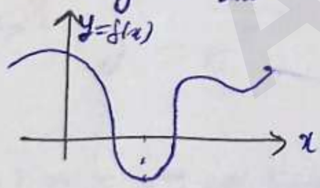
$y = (x-a)^2$

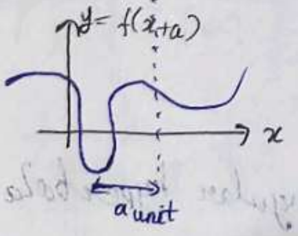
$y = (x+a)^2$











जो काम $f(x)$ मे $x=0$ पे हो रहा था। वो $f(x+a)$ मे $x=-a$ पे होगा ~~रहा था~~ और $f(x-a)$ में $x=a$ पे होगा। यानि $y=0$ पहले $x=0$ पे हो रहा था अब $f(x-a)=y$ में $y=0$ तब होगा जब $x=a$ होगा और $y=f(x+a)$ में $y=0$ तब होगा जब $x=-a$ होगा।

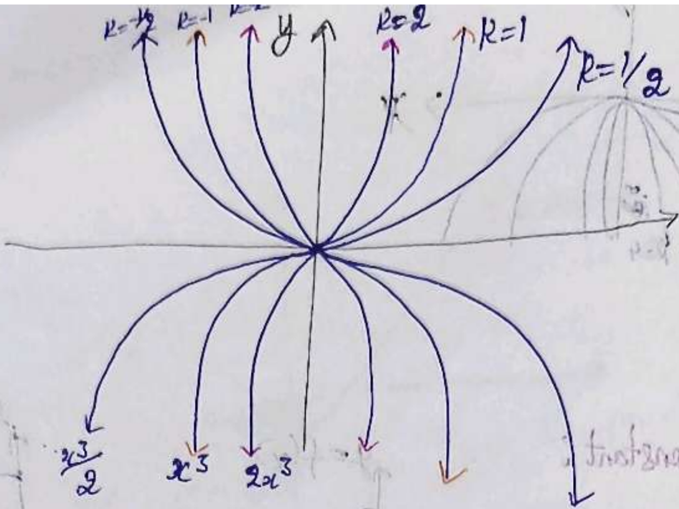
Cubic Relationship

$y \propto x^3$

$y = kx^3$

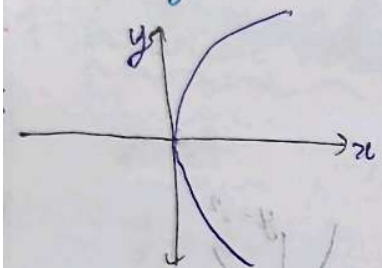
k=1

x	0	-1	1	2	-2
y	0	-1	1	8	-8



Other x^n

$x = y^2 \rightarrow x = \pm y$

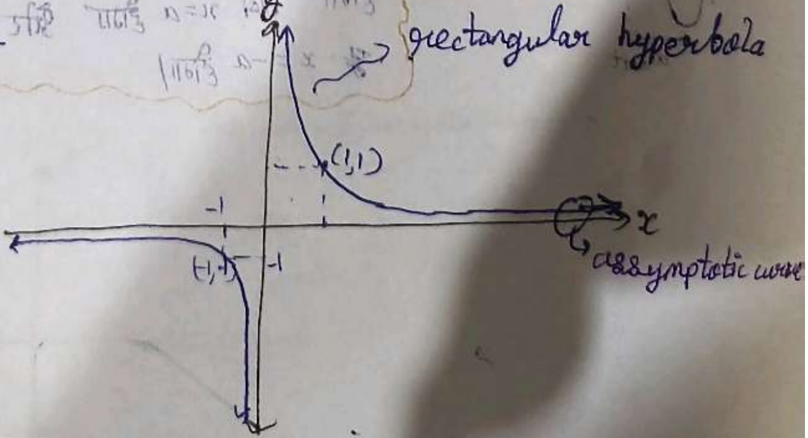


- $\boxed{y = x^{\text{even}}}$
  - Curve will lie in ~~adjustan~~ adjacent quadrate.
  - like parabola
- $\boxed{y = x^{\text{odd}}}$
  - Curve will lie in opposite quadrants.

Inverse Relationship:

$$\left.\begin{aligned} y &\propto \frac{1}{x} \\ y &= k\frac{1}{x} \end{aligned}\right\} \quad \frac{y_1}{y_2} = \frac{x_2}{x_1}$$

x	1	1/2	2	0	-2	∞	$\to 0$
y	1	2	1/2	0	-1/2	0	∞



lec-7

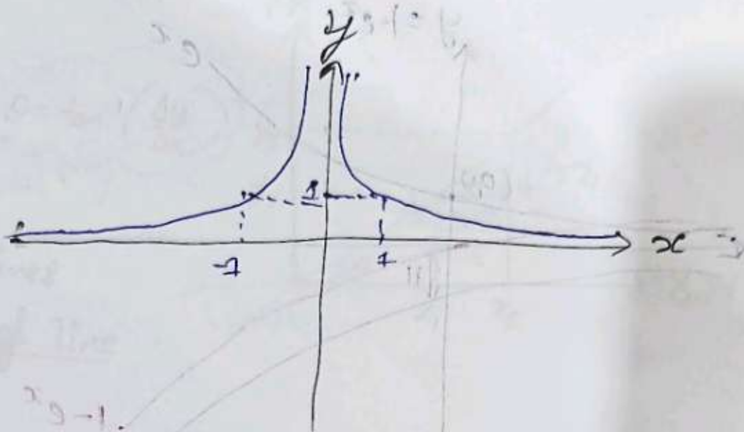
Inverse Square Relationship:

$$y \propto \frac{1}{x^2}$$

y is directly proportional to $\frac{1}{x^2}$

y is inversely proportional to x^2

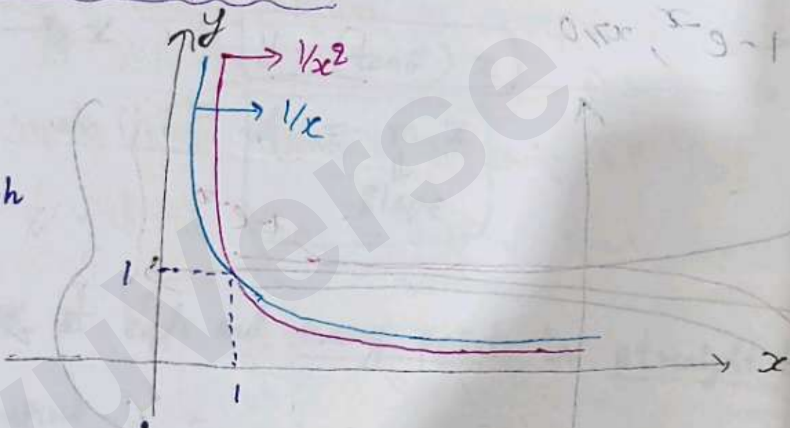
x	-0	∞	1	-1	-2	2
y	∞	-0	1	1	$1/4$	$1/4$



Compare '$\frac{1}{x}$' with '$\frac{1}{x^2}$' in I quadrant:

x	y: $1/x$	y: $1/x^2$
0	∞	∞
1	1	1
∞	0	0
2	0.5	0.25
$1/2$	2	4

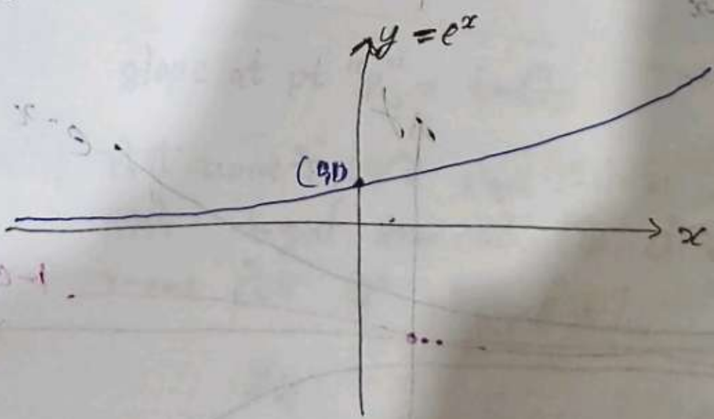
The rows for 0, 1 and ∞: same graph



Exponential relationship:

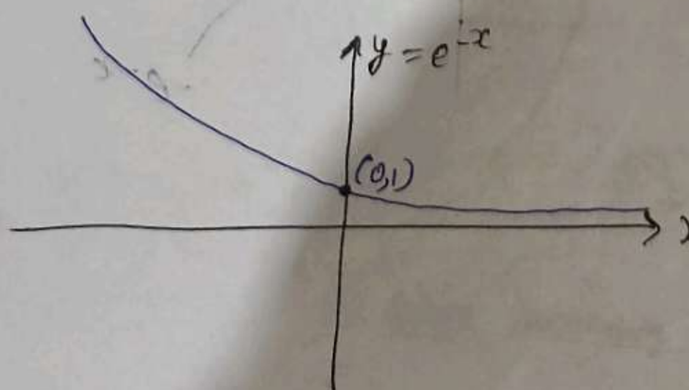
$$y = e^x$$

x	0	1	∞	$-\infty$
y	1	e	∞	0

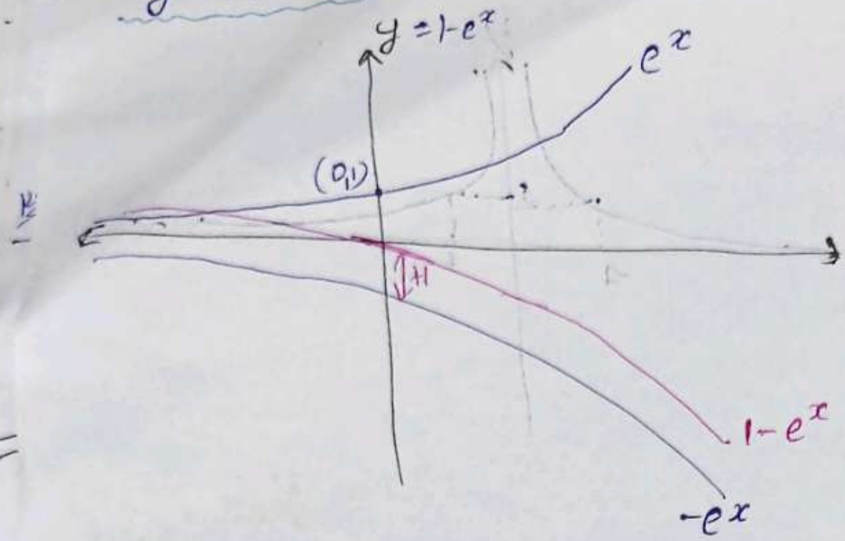


$$y = e^{-x}$$

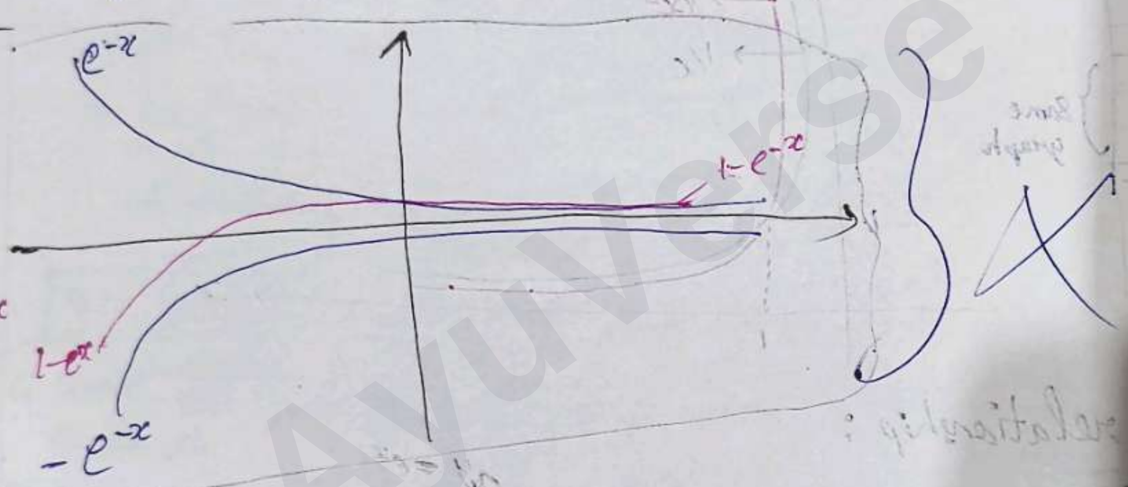
x	0	1	2	-1	-2
y	1	$1/e$	$1/e^2$	e	e^2

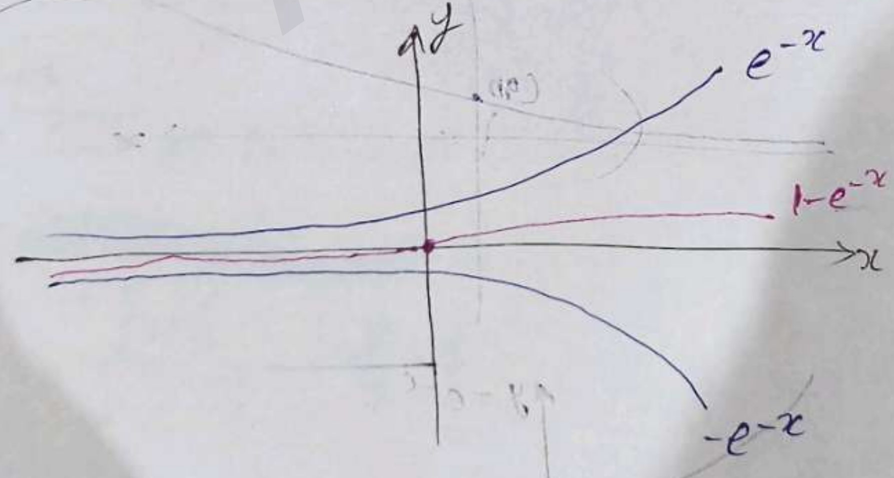


$y = 1 - e^{x},\ x > 0$



$y = 1 - e^{-x},\ x \geq 0$

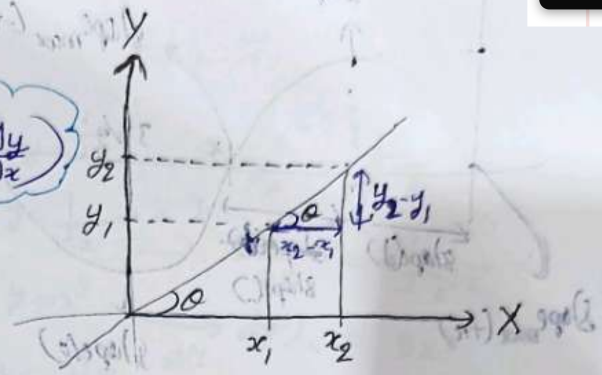




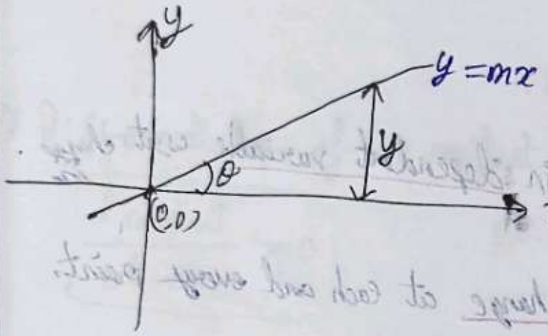
Slope Theory :

$$\boxed{\tan\theta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} \equiv \text{slope}}$$

$\theta = \tan^{-1}\left(\frac{\Delta y}{\Delta x}\right)$



- Slope of a straight line gives an idea about the inclination of line with x-axis



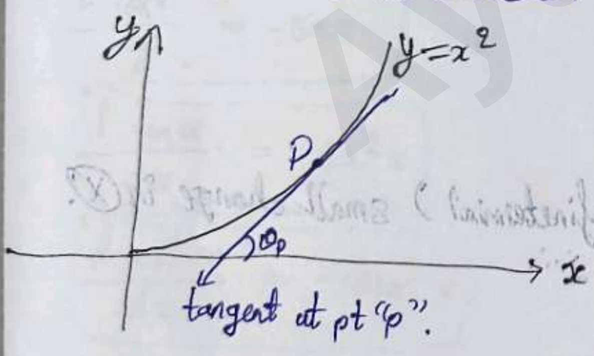
$\tan\theta = \frac{y}{x}$

$\Rightarrow \boxed{y = (\tan\theta)\, x}$

$\Rightarrow \boxed{y = m x}$ (m ⇓ slope)

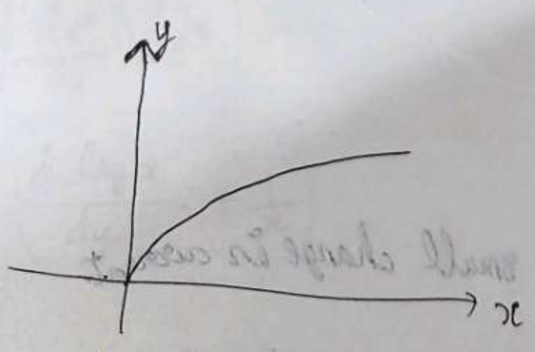
- Slope ~~is~~ will remain same at each and every point in straight line graph.

Slope at a point on curve

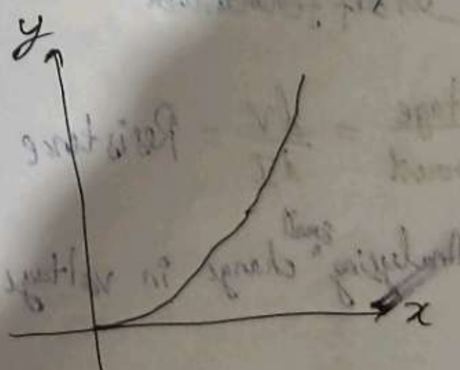


slope at pt "p" = $\tan\theta_p$

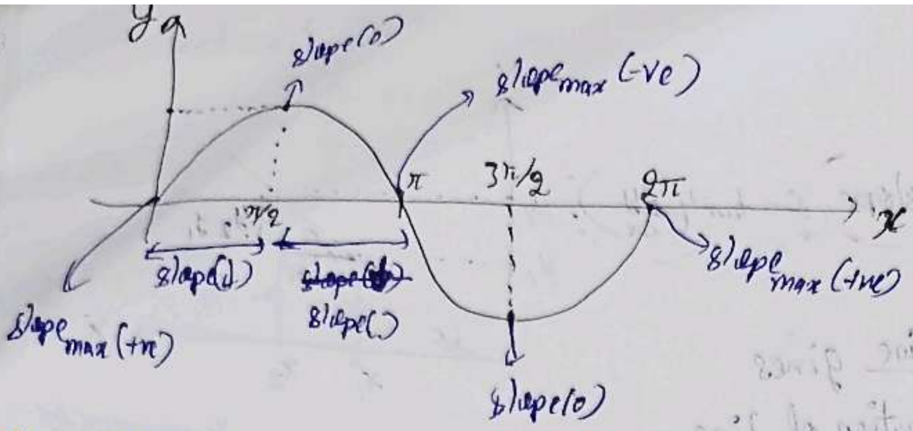
किसी curve के pt 'p' के slope निकालना है तो उसपे tangent draw करो और जो inclination x-axis होगा वही slope होगा।



slope decreasing (↓)



Slope increasing (↑)



Lec-8

Differentation:

- Differentiation is a process of analysing change in dependent variable wrt change in independent variable.
- It is a tool that gives us to analyse small change at each and every point.
- infinetesimal change in I.V → change in D.V } → analysing → Differentiation

$$\frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \left(\frac{\Delta y}{\Delta x}\right) = \boxed{\frac{dy}{dx}}$$

$\lim_{\Delta x \to 0} \Rightarrow \Delta x = 0.00000 \text{———} 1.$

$\frac{dy}{dx}$ = analysing change in (y) wrt only (infinetesimal) small change in (x).

$\frac{d\square}{dx}$ → differentiator

$$\frac{d\ \text{voltage}}{d\ \text{current}} = \frac{dV}{dI} = \text{Resistence}$$

Analysing small change in voltage wrt only small change in current.

Lec-9

Rules :-

① Power rule :

$y = x^n$

$$\frac{dy}{dx} = \boxed{\frac{d(x^n)}{dx} = n x^{n-1}}$$

Ex- $y = \sqrt{x} \xrightarrow{x^{1/2}} \frac{dy}{dx} = \frac{1}{2\sqrt{x}}$

$y = \frac{1}{\sqrt{x}} \rightarrow x^{-1/2} \rightarrow \frac{dy}{dx} = \frac{-1}{2x^{3/2}}$

$y = x^9 \longrightarrow dy/dx = 9x^8$

$y = \sqrt[3]{x} \xrightarrow{x^{1/3}} dy/dx = \frac{1}{3x^{2/3}}$

$y = \frac{1}{x} \xrightarrow[x^{-1}]{dy/dx} -1x^{-2} = -\frac{1}{x^2}$

② Differentiation of a constant:

$$\boxed{\frac{d\ \text{constant}}{dx} = 0}$$

Ex- $\frac{d2}{dx} = 0$; $\frac{d\pi}{dx} = 0$ $\frac{dG}{dx} = 0$

③ Differentiation of trignometric ration:

- $\boxed{\frac{d\sin x}{dx} = \cos x}$
- $\boxed{\frac{d\cos x}{dx} = -\sin x}$
- $\boxed{\frac{d\tan x}{dx} = \sec^2 x}$
- $\boxed{\frac{d\cot x}{dx} = -\operatorname{cosec}^2 x}$

④ $\boxed{\frac{de^x}{dy} = e^x}$

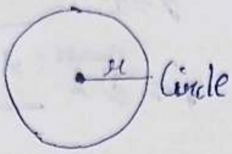
⑤ $\boxed{\frac{d\log x}{dy} = \frac{1}{x}}$

(6) Constant multiple rule :

$y = k f(x)$ ∵ k = constant

$$\boxed{\frac{dy}{dx} = k\left(\frac{df(x)}{dx}\right)}$$

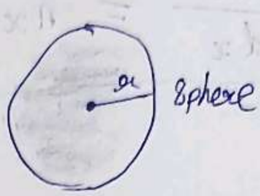
Ex- ~~$A = 4\pi r^2$~~ $A = \pi r^2$



$$\frac{dA}{dr} = \frac{d\,\pi r^2}{dr} = \boxed{\pi\ 2r}$$

$$\frac{d\text{ circumference}}{dr} = \frac{d\,2\pi r}{dr} = \boxed{2\pi}$$

$A = 4\pi r^2$



$$\frac{dA}{dr} = \frac{d\,4\pi r^2}{dr} = 4\pi\,2r = \boxed{8\pi r}$$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = \frac{d\,\frac{4}{3}\pi r^3}{dr} = \frac{4}{3}\pi\,3r^2 = \boxed{4\pi r^2}$$

Differentiation in add^n & subtraction :-

$y = f(x) \pm g(x)$

$$\boxed{\frac{dy}{dx} = \frac{df(x)}{dx} \pm \frac{dg(x)}{dx}}$$

eg :- $y = x^2 \pm \sin x$

$$\frac{dy}{dx} = \frac{dx^2}{dx} \pm \frac{d\sin x}{dx} = 2x \pm (\cos x)$$

~~$2x \mp \cos x$~~

Differentiation in product (product rule) :-

$y = f(x) \cdot g(x)$, with $f(x) = u$, $g(x) = v$

$$\boxed{\frac{dy}{dx} = u\left(\frac{dv}{dx}\right) + v\left(\frac{du}{dx}\right)}$$

eg :- $y = x^2 \cdot \sin x$

$$\frac{dy}{dx} = x^2\left(\frac{d\sin x}{dx}\right) + \sin x\left(\frac{dx^2}{dx}\right)$$

$$= x^2\cos x + \sin x(2x)$$

Division Rule (Differentiation in division) :-

$y = \frac{f(x)}{g(x)}$, $f(x) \rightsquigarrow v$, $g(x) \rightsquigarrow u$

$$\frac{dy}{dx} = \frac{u\left(\frac{dv}{dx}\right) - v\left(\frac{du}{dx}\right)}{u^2}$$

eg :- $y = $ ~~$x^2/\sin x$~~ $\sin x / x^2$

$$\frac{dy}{dx} = \frac{x^2\left(\frac{d\sin x}{dx}\right) - \sin x\left(\frac{dx^2}{dx}\right)}{x^4}$$

$$= \frac{x^2\cos x - \sin x\,(2x)}{x^4}$$

Lec-10

- $\frac{d}{dx}\left(\frac{\sin x}{\cos x}\right)$

$$= \frac{\cos x\left(\frac{d\sin x}{dx}\right) - \sin x\left(\frac{d\cos x}{dx}\right)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{\cancel{1}}{\cos^2 x} = \cancel{1}x = \boxed{\sec^2 x.}$$

- $\frac{d\, e^x \log x}{dx}$

$$= e^x\left(\frac{d\log x}{dx}\right) + \log x\left(\frac{d e^x}{dx}\right)$$

$$= \frac{e^x}{x} + \log x\, e^x = \cancel{\frac{e^x(1+\log x)}{x}}, = \boxed{\frac{e^x}{x} + (\log x)\, e^x}$$

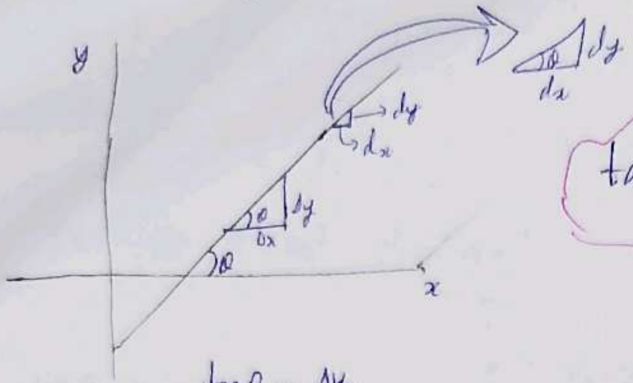
- $y = \sin 2x$

$$\frac{dy}{dx} = \frac{d\sin 2x}{dx} = \frac{d\, 2\sin x\cos x}{dx} = \cancel{2(\cos x}$$

$$= 2\,\frac{d\sin x\cos x}{dx} = 2[(-\sin^2 x) + \cos^2 x] = \cancel{2\cos^2 x - 2\sin^2 x}$$

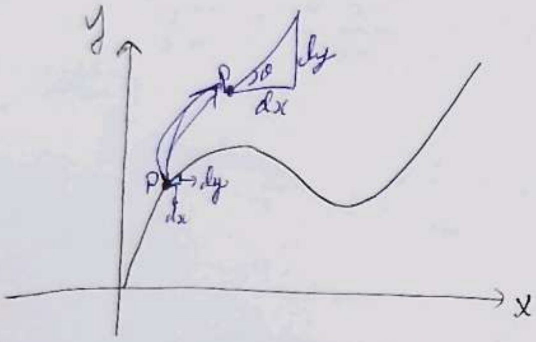
$$= 2[\cos^2 x - \sin^2 x] = \boxed{2\cos 2x.}$$

Physical meaning of differentiation:



$$\boxed{\tan\theta = \frac{dy}{dx}}$$

$$\tan\theta = \frac{\Delta y}{\Delta x}$$

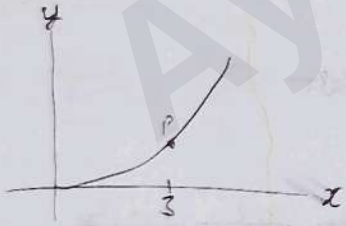


$$\tan\theta = \frac{dy}{dx}$$

$$\boxed{\text{slope at point 'P'} = \frac{dy}{dx}}$$

किसी curve पर एक particular pt. पर ~~curve~~ slope निकालना हो तो उस pt. पर $\frac{dy}{dx}$ करने पर slope मिल जाता है।

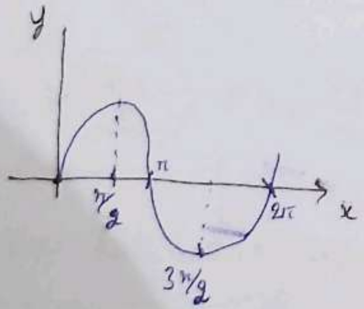
Ex →



find slope at $x = 3$ on the curve $y = x^2$

$$\text{slope}(P) = \frac{dx^2}{dx} = 2x.$$

$$\left.\frac{dy}{dx}\right|_{at\ (x=3)} = \left.2x\right|_{x=3} = \boxed{6}$$



find slope on curve $y = \sin x$ at (i) $x = 0$

(ii) $x = \pi/2$

(iii) $x = \pi$

(iv) $x = 3\pi/2$

$$\frac{dy}{dx} = \frac{d\sin x}{dx} = \cos x$$

iii) $\left.\frac{dy}{dx}\right|_{x=\pi} = \left.\cos x\right|_{x=\pi} = \boxed{-1}$

i) $\left.\frac{dy}{dx}\right|_{at\ x=0} = \left.\cos x\right|_{x=0} = \boxed{1}$

iv) $\left.\frac{dy}{dx}\right|_{at\ x=\frac{3\pi}{2}} = \cos\frac{3\pi}{2} = \boxed{0}$

ii) $\left.\frac{dy}{dx}\right|_{at\ x=\pi/2} = \left.\cos x\right|_{x=\pi/2} = \cos\frac{\pi}{2} = \boxed{0}$

Double differentiation:

differentiation ka differentiation.

$y = f(x)$

↳ $\frac{dy}{dx}$ = analysing change in y w.r.t change in x ≡ slope

↳ $\frac{d\left(\frac{dy}{dx}\right)}{dx}$ = analysing change in slope w.r.t change in x

Eg: $\frac{dx}{dt}$ → velocity (v)

↳ $\frac{d\left(\frac{dx}{dt}\right)}{dt}$ = acceleration (a).

$y \xrightarrow[\frac{dy}{dx}]{1^{st}} \frac{dy}{dx} \; (y') \xrightarrow[\frac{d}{dx}]{2^{nd}} \frac{d\left(\frac{dy}{dx}\right)}{dx} = \frac{d^2y}{dx^2} \; (y'')$

↳ 2nd derivative of

$\frac{dy}{dt} = \dot{y}$

where, t is time.

$\frac{d^2y}{dt^2} = \ddot{y}$

Chain Rule: जब fun^c के अंदर mtlb fun^c के ऊपर एक फिर fun^c हो कहा chain rule.

$x \rightarrow$ [Fun^c 1: Square, $f(x)$] $\xrightarrow{x^2}$ [Fun^c 2: Sin, $g(f(x))$] $\xrightarrow{\sin x^2}$ [Fun^c 3: exponential, $h(g(f(x)))$] $\rightarrow e^{\sin x^2}$

$y = g(f(x))$

$y = h(g(f(x)))$

Eg :-
- $\sin(Ax)$
- $(Ax+B)^2$
- $\sin(x^2)$
- (e^{Ax+B})
- $\frac{1}{(Ax+B)}$

जैसे प्याज को छीलते वक्त सबसे ऊपर का हटाते हैं फिर उसके अंदर वाला वैसे ही chain rule है।

Ex → $y = e^{\sin(x^2)}$

$$\frac{dy}{dx} = \frac{d e^{(\alpha)}}{dx} \times \frac{d \sin(\beta)}{dx} \times \frac{d x^2}{dx}$$

$$= \boxed{e^{\sin x^2} \cdot \cos x^2 \cdot 2x}$$

$y = Duda(Papa(Beta))$

$$\frac{dy}{dx} = \underbrace{\frac{d\ Dada}{dx}}_{\substack{(Papa\ Beta) \\ Dummy}} \times \underbrace{\frac{d\ Papa}{dx}}_{\substack{(Beta) \\ Dummy}} \times \frac{d\ Beta}{dx}$$

$y = \sin(Ax+B)$, where A & B constant

$$\frac{dy}{dx} = \frac{d \sin(Ax+B)}{dx} \times \frac{d(Ax+B)}{dx}$$

$$= \cos(Ax+B) \times \left[\frac{d\,Ax}{dx} + \frac{dB}{dx}\right]$$

$$= \boxed{\cos(Ax+B)\, A}$$

Lec-11

Q) Radius of sphere $= 0.5t^2$ {t = time}. Volume of sphere changing with time. Find rate of change of volume when (t=4)

Soln:- $V = \frac{4}{3}\pi r^3$

method ① $\frac{dV}{dt} = \frac{d\,\frac{4}{3}\pi r^3}{dt}$

$= \frac{d\,\frac{4}{3}\pi(0.5t^2)^3}{dt} = \frac{4}{3}\pi(0.5)^3\frac{d\,t^6}{dt}$

$= \frac{4}{3}\pi(0.5)^3\, 6t^5 = \boxed{8\pi(0.5)^3 t^5\Big|_{t=4}}$

method ② $\frac{dV}{dt} = \frac{d\,\frac{4}{3}\pi r^3}{dt}$

$= \frac{4}{3}\pi\, 3r^2 \times \frac{dr}{dt} = \frac{4}{3}\pi\, 3r^2 \times \frac{0.5t^2}{2}$

$= 4\pi(0.5t^2)^2\, 0.5(2t)$

$= \boxed{4\pi(0.5t^2)^2\, 0.5(2t)\Big|_{t=4}}$

life style → Papa salary → month

(L) (S) (M)

$L = S^2 + 2S + 6$; $S = -M^2 + 2M$

$\frac{dL}{dM}$ or $\frac{dL}{dS} \times \frac{dS}{dM}$

$y = ax^2$ & $x = \sqrt{t}$

$$\frac{dy}{dx} = a2x = 2ax.$$

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} = \frac{d\,ax^2}{dx} \times \frac{d\sqrt{t}}{dt} = 2ax \times \frac{1}{2\sqrt{t}} = a\sqrt{t} \times \frac{1}{\sqrt{t}} = \boxed{a}.$$

or

$$\frac{dy}{dt} = \frac{d\,ax^2}{dt} = \frac{2a\,d(t)}{dt} = \cancel{2} \qquad \frac{dy}{dt} = \frac{d\,at}{dt} = \boxed{a}$$

Q-) Velocity as a func^n of position (x)

$$v = -x^2 + 2x + 6$$

& pos^n as a func^n of time (t). $(x = 1 + 2t)$

Find $\frac{dv}{dt} = ?$

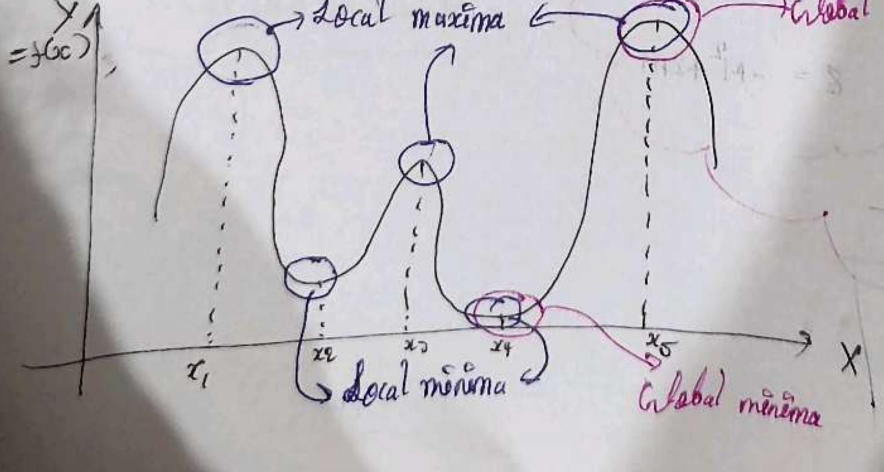
$$\frac{dv}{dt} = \frac{d(-x^2+2x+6)}{dx} \times \frac{d(1+2t)}{dt} = (-2x+2)(2)$$

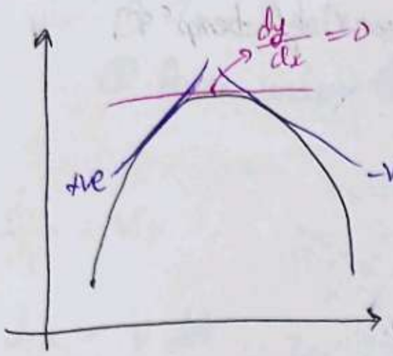
$$= \boxed{4 - 4x.}$$

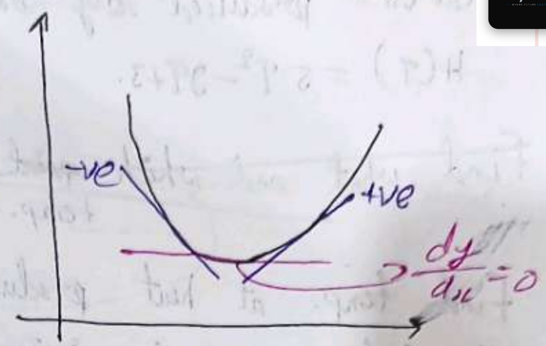
$$= 4 - 4(1+2t)$$

$$= \boxed{8t} \text{ Ans}$$

Maxima & Minima :







$$\left.\frac{d\,slope}{dx}\right|_{\frac{dy}{dx}=0} = \frac{(-ve)-(+ve)}{dx} = -ve$$

$$\left.\frac{d\,slope}{dx}\right|_{\frac{dy}{dx}=0} = \frac{+ve-(-ve)}{dx} = +ve$$

① जिस func की maxima, minima निकालनी हो, उस funn का 1st differentiation जिन pt. पर '0' आता है, वो pt. possibility होते है maxima और minima के लिए। (slope is 0 where maxima and minima may occur).

② कहा maxima होगा कहा Minima इस बात को मै 2nd derivative slope 0 वाले pt. पर निकालकर conform कर लेता।

Lec-12

Q) No. of commuters $= t^3-3t^2+6$, where 't' is time.
Find time when N is maximum & minimum also find maximum / minimum value of "N"! {for range $-1<t<3$}

Soln:- $N = t^3-3t^2+6$

$$\frac{dN}{dt} = 3t^2-6t$$

$$\Rightarrow 3t^2-6t=0 \Rightarrow 3t(t-2)=0 \Rightarrow t=0,\ t=2.$$

~~Maxima~~ $\frac{d^2N}{dt^2} = 6t-6$

$6t-6\big|_{t=0} = -6$; $6t-6\big|_{t=2} = 6$.

∴ Maxima is at 't=0'.
Minima is at 't=2'.

$N_{max} = (t^3-3t^2+6)\big|_{t=0}$
$= \boxed{6}$

$N_{min} = (t^3-3t^2+6)\big|_{t=2}$
$= \boxed{2}$

Q.) Heat (H) produced by body is defined in terms of temp 'T';
$H(T) = 5T^2 - 9T + 3$.

~~Find what and which point is maxima / minima.~~ temp.

Find temp. at heat produced is maximum / minimum.
Also find maximum & minimum heat.

Solⁿ :- $\frac{d(H)}{dT} = 10T - 9$

$\Rightarrow 10T - 9 = 0$

$\Rightarrow \boxed{T = 9/10}$ → means $\left(T = \frac{9}{10}\right)$ is the possible temp. where max or min may occur.

$\frac{d^2H}{dT} = 10 > 0$

→ at $T = \frac{9}{10}$ minima will occur.

$$H_{min}\Big|_{t=\frac{9}{10}} = 5\left(\frac{81}{100}\right) - \frac{81}{10} + 3 = \frac{405 - 810 + 300}{100} = -\frac{105}{200} = \boxed{-\frac{21}{20}}$$

Ans

INTEGRATION

$$\sum_{i=1}^{n} dx_i = \int dx$$

- It is a process of summation or reverse process of differentiation.
- It is also called (anti derivative).

Note: To add discrete values we use 'Σ'.

Ex → $\sum_{i=1}^{50} M_i = M_1 + M_2 + M_3 + \text{------} + M_{50}$.

- A process of addition of infinitesimal, ~~contin~~ continuous changes is known as integration.

Application: ① To calculate net change in physical quantity.
② Area under the curve.

$\frac{dx}{dt} = v$

$dx = v\,dt$

only small change in pos^n in time "dt".

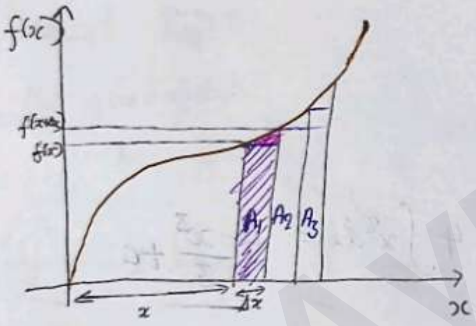
$\overset{dx_1}{\underset{dt}{\longleftrightarrow}}\ \overset{dx_2}{\underset{dt}{\longleftrightarrow}}\ \overset{dx_3}{\underset{dt}{\longleftrightarrow}}$ - - - - - - → ∞ 'dt' interval.

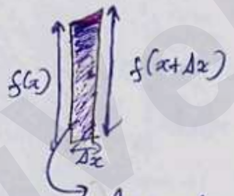
$\sum_{i=1}^{\infty} dx_i \Rightarrow \int dx$ --→ integration of dx. gives net change in pos^n Δx

$\boxed{\int v\,dt}$ → integrating velocity wrt time.

$\int dx \rightarrow \Delta x$

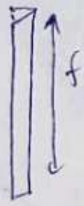
#





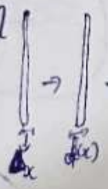
Area of strip $= f(x)\,\Delta x = A_1$

$A_1 + A_2 + A_3 + \ldots$ error, bcz area missing



$f(x+\Delta x) \rightarrow$ Area $= f(x+\Delta x)\,\Delta x$ — Extra area come error

∵ if Δx is ~~app~~ very very small
∴ $f(x) \approx f(x+\Delta x)$



→ Area of strip $= f(x)\,dx$

∞ no. of strips:

Total strip $= \int dA = \int f(x)\,dx$

Mathematical formula

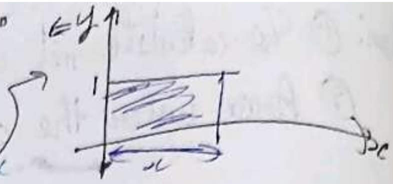
1) Power rule :

$$\boxed{\int x^n dx = \frac{x^{n+1}}{n+1} + C}$$

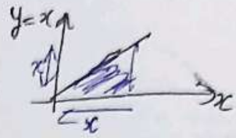
except

$$\boxed{\int \frac{1}{x} dx = \ln x + C}$$

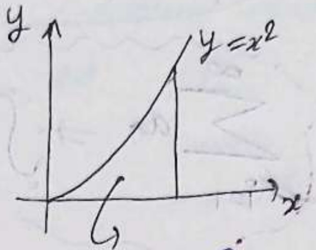
eg → ① $\int x^0 dx = x + C$



② $\int x^1 dx = \frac{x^2}{2} + C$



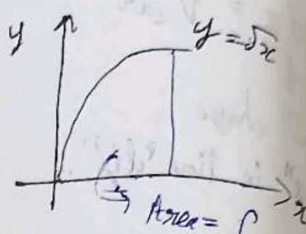
③



Area $= \int x^2 dx$

$= \frac{x^3}{3} + C$

④



Area $= \int \sqrt{x}\, dx$

$= \frac{x^{3/2}}{3/2}$

$= \frac{2x^{3/2}}{3} + C$

② $\int \sin x\, dx = -\cos x + C$

~~$\int \cos$~~ $\int \cos x\, dx = \sin x + C$

③ $\int e^x dx = e^x + C$

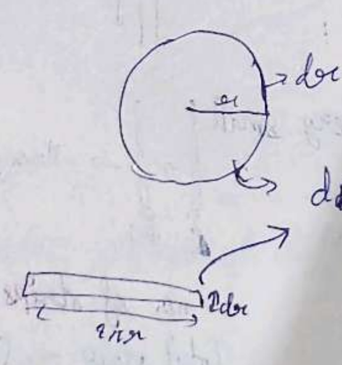
④ $\int (k f(x))\, dx = k \int f(x)\, dx + C$ → eg → $\int 4x^2 dx = 4 \int x^2 dx = 4 \frac{x^3}{3} + C$

Learn

⑤ ~~Consta~~

eg → $\int 2\pi r\, dr$

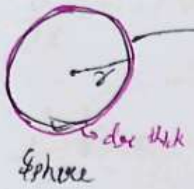
$= 2\pi \times \frac{r^2}{2} + C = \pi r^2 + C$



$dA = (2\pi r)\, dr$

$\int dA = \int 2\pi r\, dr$

$= 2\pi \times \frac{r^2}{2} = \pi r^2 + C$



$$dv = (4\pi r^2)\,dr$$

$$\int dv = \int (4\pi r^2)\,dr$$

$$= 4\pi \times \frac{r^3}{3} + c$$

$$= \frac{4\pi}{3} r^3 + c$$

(5) Addition & Substraction :

$$y = f(x) \pm g(x)$$

$$\boxed{\int y\, d(x) = \int (f(x) \pm g(x))\,dx = \int f(x)\,dx \pm \int g(x)\,dx}$$

eg :- $y = x^2 \pm \sin x$

$$\int y\, d(x) = \int (x^2 \pm \sin x)\,dx$$

$$= \frac{x^3}{3} \pm (-\cos x) + c$$

$$= \frac{x^3}{3} \mp \cos x + c$$

(6) Product Rule:

$$y = \underset{\mathrm{I}}{f(x)}\, \underset{\mathrm{II}}{g(x)}$$

$$\int y\, dx = \mathrm{I} \int \mathrm{II}\, dx - \int \left(\frac{d\mathrm{I}}{dx} \int \mathrm{II}\, dx \right) dx$$

$$\cancel{y = \mathrm{I} \int \mathrm{II}\, dx - \int \left(\frac{d\mathrm{I}}{dx} \int \frac{d\mathrm{II}}{dx} \right) dx}$$

I L A T E

I → Inv. trigo. func

L → Log

A → Algebraic

T → Trigo.

E → Exponent

$$y = \underset{\mathrm{I}}{\sin x}\ \underset{\mathrm{II}}{e^x}$$

$$y = \underset{\mathrm{II}}{x^2} \cdot \underset{\mathrm{I}}{\log_e x}$$

Example :

i) $y = \underset{\mathrm{I}}{x} \cdot \underset{\mathrm{II}}{e^x}$

$$\int y\, d(x) = x \int e^x\, d(x) - \int \left(\overset{1}{\frac{dx}{dx}} \int e^x\, dx \right) dx$$

$$= \boxed{x \cdot e^x - e^x + c}$$

$x \xrightarrow{} \boxed{\text{Multiplier } (A)} \xrightarrow{Ax} \boxed{\sin} \longrightarrow \sin(Ax)$

$$\int \text{Papa}(\text{Beta})\,dx = \frac{\int(\text{Papa})\,dx}{\frac{d(\text{Beta})}{dx}}$$

Beta → linear fun'

eg ⓪ $\int \sin(Ax)\,dx = \frac{-\cos(Ax)}{\frac{d(Ax)}{dx}} = \frac{-\cos(Ax)}{A} + C$

⓪ $y = (Ax+b)^2$

$$\int y\,d(x) = \int (Ax+b)^2\,dx$$

$$= \frac{\frac{(Ax+b)^3}{3}}{A} = \frac{(Ax+b)^3}{3A} + c$$

⓪ $y = \sin(\omega t + \theta)$

$$\int y\,dt = \frac{-\cos(\omega t+\theta)}{\frac{d(\omega t+\theta)}{dt}} = \frac{-\cos(\omega t+\theta)}{\omega} + c$$

Short

Differentiation	Integration
1) $\frac{d}{dx}x^n = nx^{n-1}$	1.) $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \;;\; n \neq -1$
2.) $\frac{dc}{dx} = 0$	2.) $\int c\,dx = cx' + C$
3.) $\frac{d}{dx}kx^2 = k\frac{dx^2}{dx}$	3.) $\int kx^2\,dx = k\int x^2\,dx$
4) $\frac{d}{dx}e^x = e^x$	4.) $\int e^x\,dx = e^x + C$
5) $\frac{d}{dx}a^x = a^x \ln a$	5.) $\int a^x\,dx = \frac{a^x}{\ln a} + C$

6.) $\frac{d}{dx}(\ln x) = \frac{1}{x}$

6.) $\int \frac{1}{x}\, dx = \ln x$

7.) $\frac{d}{dx}\sin x = \cos x$

7.) $\int \sin x\, dx = -\cos x + C$

8.) $\frac{d}{dx}\cos x = -\sin x$

$\int \cos x\, dx = \sin x + C$

9.) $\frac{d}{dx}\tan x = \sec^2$

10.) $\frac{d}{dx}\cot x = -\operatorname{cosec}^2$

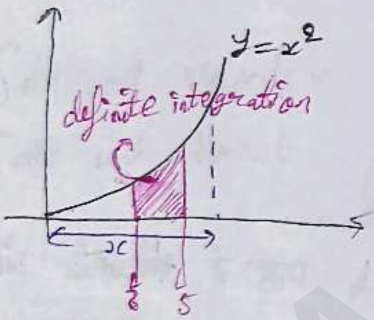
11.) $\frac{d \log_{10} x}{dx} = \frac{1}{x \ln x}$

Lec-15

Definite Integration:

$$\int_a^b f(x)\, dx = \Big|g(x)\Big|_a^b = g(b) - g(a)$$

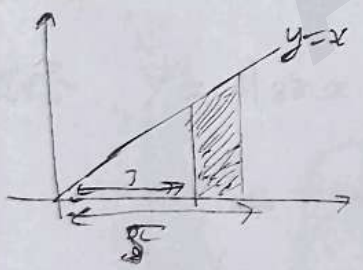
$|\;|$ → This is not mod, it is symbol

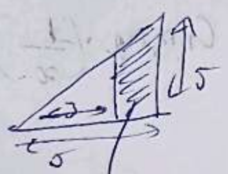


$$\int x^2\, dx = \frac{x^3}{3} + C$$

$$\int_{x=3}^{x=5} y\, dx = \int_3^5 x^2\, dx = \left|\frac{x^3}{3}\right|_3^5 = \frac{125}{3} - \frac{27}{3}$$

$$= \frac{98}{3} \text{ sq unit}$$





Area $= \frac{25}{2} - \frac{9}{2} = \frac{16}{2} = 8$ sq unit

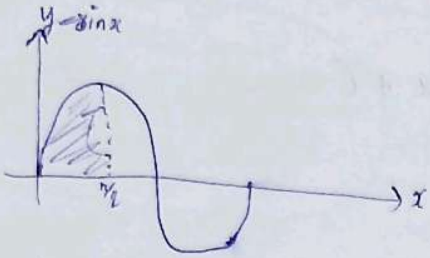
$$\int_{x=3}^{x=5} y\, dx = \int_3^5 x\, dx = \left|\frac{x^2}{2}\right|_{x=3}^{x=5}$$

$$= \frac{25}{2} - \frac{3^2}{2} = \frac{16}{2} = 8 \text{ sq unit}$$

definite integration numerical value देता है, तो उसमें const. add नहीं करते वहीं indefinite में const. C add करते हैं।

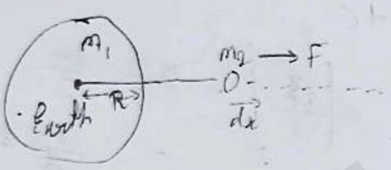
Q4 find area :-

i.) Btw 0^0 to $\frac{\pi}{2}$ in sin x graph



$$\int_0^{90^\circ} \sin x \, d(x) = \left| -\cos x \right|_0^{90} = -0-(-1) = \boxed{+1} \text{ Ans}$$

Q4 find $\int F \cdot dx$



$$\int_R^{\infty} F \cdot dx = \int_R^{\infty} \frac{G m_1 m_2}{x^2} \, dx = G m_1 m_2 \cdot \left(\frac{-1}{x}\right) \Bigg|_R^{\infty}$$

$$= 0 - G m_1 m_2 \left(\frac{-1}{R}\right)$$

$$= \frac{G m_1 m_2}{R}$$

$\frac{1}{x} = x^2$

	$\frac{dy}{dx}$	$\int dx$
① $y = x^2 + 1 \longrightarrow$	$2x \longrightarrow$	$x^2 + C$
② $y = x^2 + 8 \longrightarrow$	$2x \longrightarrow$	$x^2 + C$
③ $y = x^2 + 3 \longrightarrow$	$2x \longrightarrow$	$x^2 + C$

① $y(x=0) = 1$

② $y(x=0) = 8$

③ $y(x=0) = 3$

indf. integration करते पर एक चीज miss हो जाती है, y के लिए x=0 miss हो जाती है। इस लिए const. C add होता है।

$a = \frac{dv}{dt}$

$dv = a\,dt$

$\int dv = \int a\,dt$

$\int_{v_i}^{v_f} dv = a\int_0^t dt$

$|v|_{v_i}^{v_f} = a|t|_0^t$

$\Rightarrow v_f - v_i = at$

$\Rightarrow v_f = v_i + at$

$\int dv$ wrt velocity v — $v_f - v_i$

$\int a\,dt$ wrt time t — $a\int dt$, $\hookrightarrow at$

$v = \frac{dx}{dt}$

$dx = v\,dt$

$\int_{x_i}^{x_f} dx = \int_0^t v\,dt$

$\Rightarrow |x|_{x_i}^{x_f} = \int_0^t (u+at)\,dt$

$\Rightarrow x_f - x_i = \int_0^t u\,dt + \int_0^t at\,dt$

$\Rightarrow x_f - x_i = ut + a\frac{t^2}{2}$

$\Rightarrow x_f = ut + \frac{1}{2}at^2 + x_i$

$\int dx$ wrt position x — $x_f - x_i$

$\int dt$ wrt time t — t

$\int t\,dt$ wrt time t — $\frac{t^2}{2}$

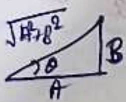
$\therefore$ जिस variable के साथ d लगा हो, integration उसी के respect me hota hai.

Lec-15

Note:- $y = A\sin x \pm B\cos x$

$\hookrightarrow y_{max} = +\sqrt{A^2+B^2}$

$\hookrightarrow y_{min} = -\sqrt{A^2+B^2}$



Proof $y = \sqrt{A^2+B^2}\left(\frac{A}{\sqrt{A^2+B^2}}\sin x + \frac{B}{\sqrt{A^2+B^2}}\cos x\right)$

$\Rightarrow y = \sqrt{A^2+B^2}\,(\cos\theta\sin x + \sin\theta\cos x)$

$\Rightarrow y = \sqrt{A^2+B^2}\,\underbrace{(\sin(\theta+x))}_{\text{sin का fun}^c}$

$\therefore -\sqrt{A^2+B^2} \le y \le \sqrt{A^2+B^2}$

Binomial & approximation

$(1+x)^2 = 1+2x+x^2$

$(1+x)^3 = 1+x^3+3x^2+3x$

$(1+x)^4 = 1+4x + \cancel{5}x^2 + 4x^3 + x^4$

⋮

$(1+x)^{1/2}$ or $(1+x)^{1/3}$

$(1+x)^{2/3}$ — ? ?

condition $x <<< 1$ ⇝ $x^3, x^2, 5x^2, 4x^2 \dots \approx 0$.

$\boxed{(1\pm x)^n \approx 1 \pm nx}$ → Binomial

Ex → $(1003)^2 = (1000+3)^2$

$= (1000)^2\left[1+\frac{3}{1000}\right]^2$

$= 10^6\left[1+2(0.003)\right]$

Arithmatic Progression (A.P)

a_1 1st term	a_2 2nd term	a_3 3rd term	....	a_n n^{th} term
a	$a+d$	$a+2d$		$a+(n-1)d$

d = common difference

Sum of n terms of Arithmatic progressive series ⇒ S_n

$$S_n = \frac{n}{2}(a+a_n) = \frac{n}{2}(2a+(n-1)d)$$

Geometric Progression (G.P)

a_1 1st term	a_2 2nd term	a_3 3rd term	....	a_n n^{th} term
a	ar	ar^2		ar^{n-1}

r = common ratio

Sum of n terms of geometric progressive series = S_n

$$S_n = \frac{a(r^n-1)}{r-1} = \frac{a(1-r^n)}{1-r}$$

when $r < 1$ infinite G.P

$$S_n = \frac{a}{1-r}$$

Eg ~

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \text{ - - - - - - - - } \infty$$

$$S_n = \frac{1}{1-\frac{1}{2}} = 2.$$

$$1 + \frac{1}{e} + \frac{1}{e^2} + \frac{1}{e^3} + \text{ - - - - - - - - } \infty$$

$$S_n = \frac{1}{1-\frac{1}{e}} = \frac{e}{e-1}.$$

lec-

VECTORS

Physical Qut:- Those quantity which can be measure and have some physical significance
Eg:- mass, area, force, etc.

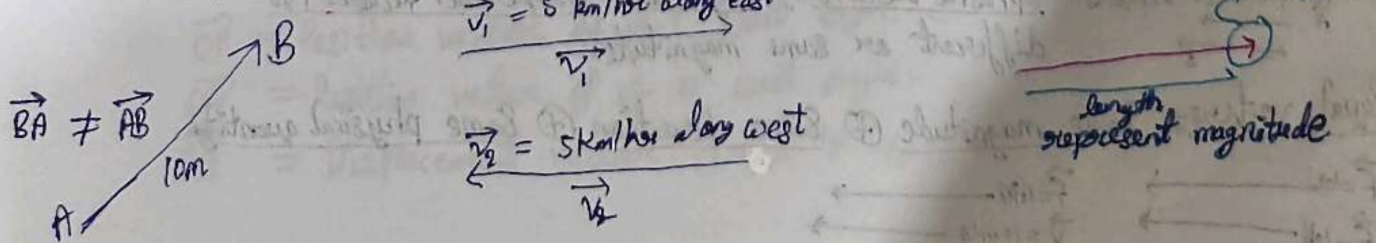
→ Scalar Quantity:- Those physical quantity which have not direction as significance but magnitude should significant.
(magnitude)
Ex- mass, area, etc.

→ Vector Quantity:- Those physical quantity which have significance of magnitude along with direction.
(magnitude + dirn)
Ex- force, velocity, etc

Representation of vectors:

① Arrow representation
② Magnitude representation
③ Cartesian form

Arrow representation

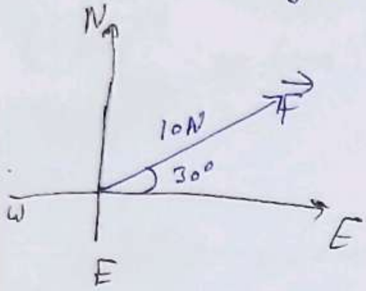


Vector = Magnitude + direction

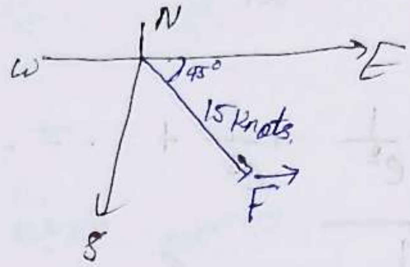
$\vec{A} = |\vec{A}| \cdot \hat{A}$

Magnitude Representation

$\vec{F} = 10N$ 30° N of E

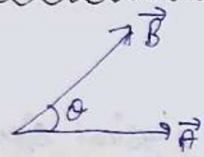


wind velocity $(\vec{V}_w) = 15$ knots (SE)

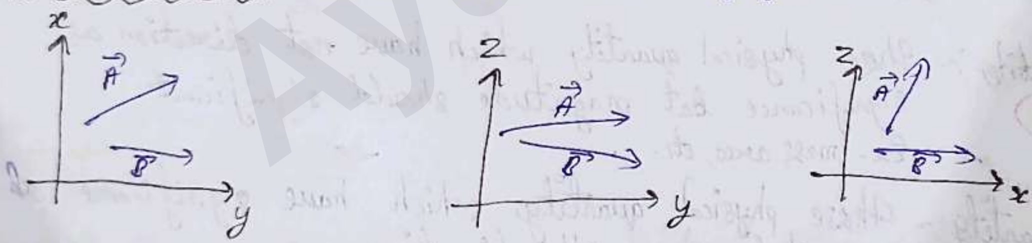


Types of Vectors:

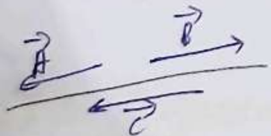
① Co-initial vectors - Two or more vectors having same initial point.



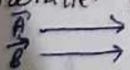
② Co-planar vectors - Two or more vectors lying in the same plane.



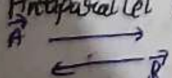
③ Co-linear vectors - Two or more vectors lying along the same line.



* Parallel vectors - same dirn; angle b/w parallel vectors $= 0^\circ$
  different or same magnitude.



* Antiparallel vectors - opposite direction; angle b/w antiparallel vector $= 180^\circ$
  different or same magnitude.



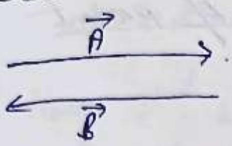
④ Equal vectors - Same magnitude + Same direction + Same physical quantity

$\vec{F} = 10N \longrightarrow$ $\vec{F} = 10N \longrightarrow$

$\vec{F} = 10N \longrightarrow$ $\vec{V} = 10 m/s \longrightarrow$

(✓) (X)

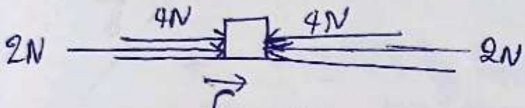
⑤ Anti vector or negative vector : Dir^n of a vector is made by rotating 180° another vector.



$$\left.\begin{aligned}\vec{B} &= -\vec{A}\\ \vec{A} &= -\vec{B}\end{aligned}\right\}$$ $\vec{B}$ vector is antiparallel of $\vec{A}$ vector or vice-versa

$|\vec{A}| = |\vec{B}|$

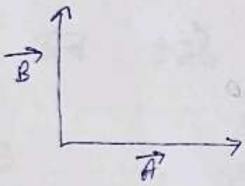
⑥ Null vector : A vector with 0 magnitude.
Denoted by : • (dot)
direction : ∞



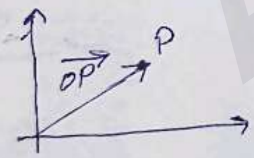
$\vec{F}_{resultant} = 0\ N$

Lec-18

⑦ Orthogonal Vector : Two vector Perpendicular to each other.

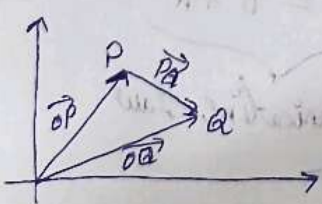


⑧ Position Vector :



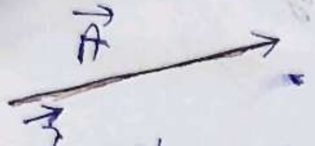
$\vec{OP}$ = Position^n vector of pt 'P' wrt 'origin'.

⑨ Displacement Vector :



$\vec{OP}$ = Position vector of pt. 'P' wrt origin.
$\vec{OQ}$ = Position vector of pt. 'Q' wrt origin.
$\vec{PQ}$ = Displacement vector from 'P' to 'Q'.

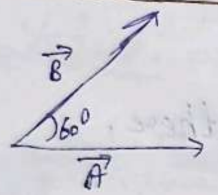
⑩ Unit Vector : A vector with magnitude 1 but have direction of a parent vector means represent direction of parent vector.

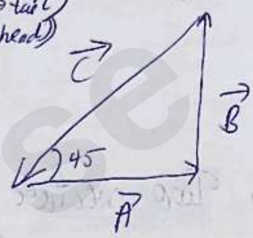


unit magnitude wala $\vec{A}$ ki dir^n mein → $\hat{A}$

Eg : $\hat{i}$ – unit vector along x axis
$\hat{j}$ – 〃 〃 y axis
$\hat{k}$ – 〃 〃 z axis

Angle between two vector : (join them from tail → tail or head → head)





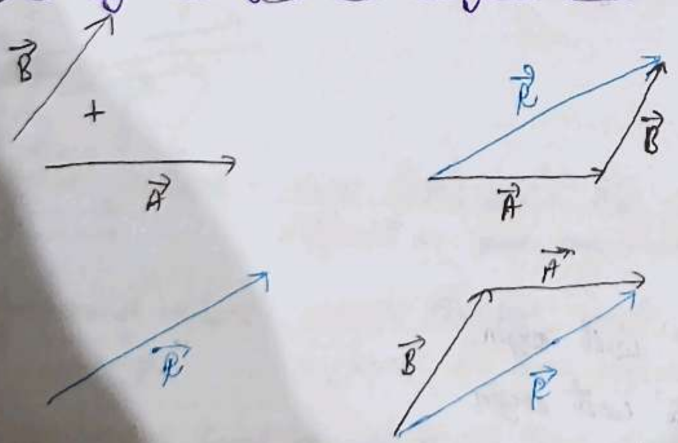
Angle b/w $\vec{A}$ & $\vec{B}$ $= \vec{A} \wedge \vec{B} = 60^0$

$\vec{A} \wedge \vec{B} = 90^0$
$\vec{B} \wedge \vec{C} = 135^0$
$\vec{C} \wedge \vec{A} = 135^0$

Vector law of Addition :

① Law of triangle
② Law of parallelogram
③ Polygon law

Law of triangle / Δ law of addition : used to add two vectors

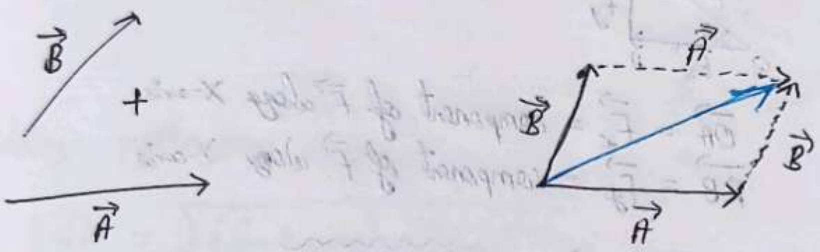


$$\vec{R} = \underbrace{\vec{A} + \vec{B} = \vec{B} + \vec{A}}_{\text{Communicative law}}$$

Step① जिन दोनो vector को add करना हो, उनको same sense में arrange such that the two vector represent the adjacent sides of triangle

Step② Now, 3rd side of triangle in opposite sense (from tail of first vector to head of second vector.), will give you the resultant vector.

Parallelogram law of addition: Used to add two vectors only



Step① जिन दो vector को add करना है, उन्हे co-initial sense में arrange करे such that they represent the adjacent side of parallelogram.

Step② अब इनके बिच का diagonal resultant vector होगा/

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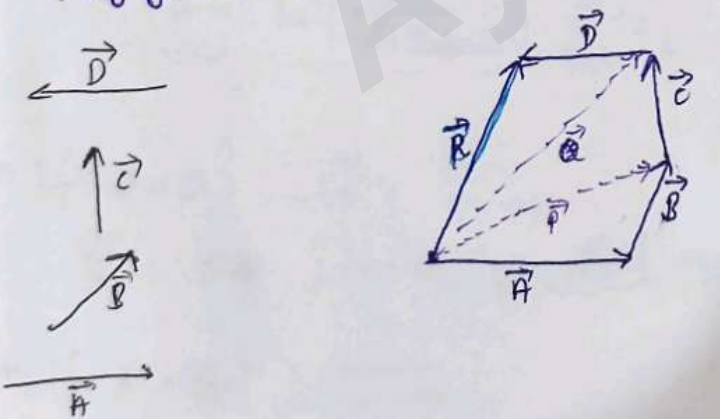
Polygon Law: used to add multiple vectors



$\vec{P} = \vec{A} + \vec{B}$

$\vec{Q} = \vec{P} + \vec{C}$

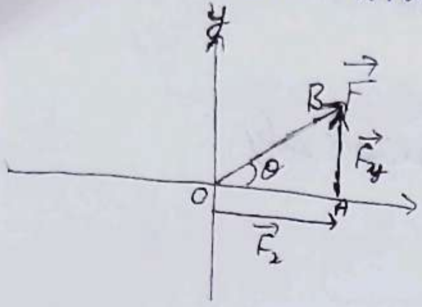
$\vec{R} = \vec{Q} + \vec{D}$

$\vec{R} = \vec{P} + \vec{C} + \cancel{\vec{A}+\vec{B}}\ \vec{D}$

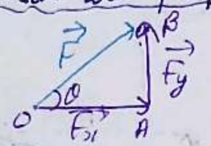
$\vec{R} = \vec{A} + \vec{C} + \vec{B} + \vec{D}$

Resolution of vectors :

किसी भी vector को 'n' number of independent vector में split/resolve कर सकते हैं और उनके sum वापस parent vector देंगे। Ex $\vec{F} = \vec{F_1} + \vec{F_2} + \dots \vec{F_n}$



A trigonometry method of resolution:
→ splitting of a vector into two vectors which are perpendicular to each other → rectangular resolution



$\vec{OA} = \vec{F_x}$ = component of $\vec{F}$ along X-axis
$\vec{AB} = \vec{F_y}$ - component of $\vec{F}$ along Y-axis

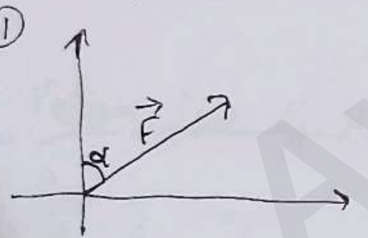
$\vec{F} = F\sin\theta + F\cos\theta$
$\vec{F} = F\sin\theta\,\hat{j} + F\cos\theta\,\hat{i}$

⊙ $\sin\theta = \frac{AB}{OB} = \frac{\vec{F_y}}{\vec{F}} \Rightarrow \vec{F_y} = F\sin\theta$

⊙ $\cos\theta = \frac{OA}{OB} = \frac{\vec{F_x}}{\vec{F}} \Rightarrow \vec{F_x} = F\cos\theta$

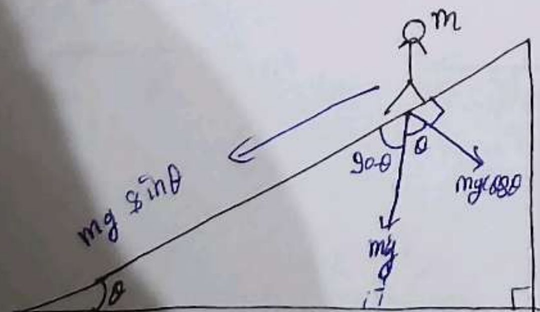
Examples :

①



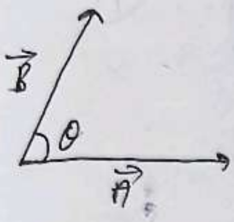
$\vec{F} = F\sin\theta\,\hat{i} + F\cos\theta\,\hat{j}$

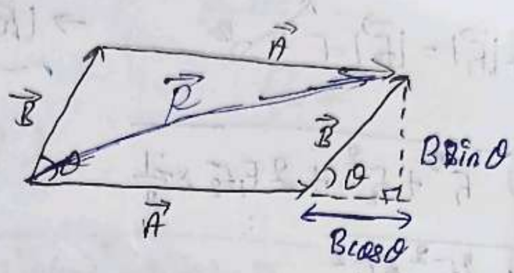
②

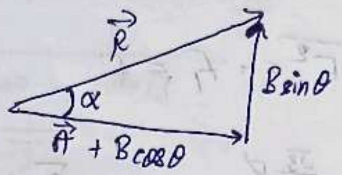


- $mg\cos\theta$ → Perpendicular resolution to the surface responsible for making contact with surface.
- $mg\sin\theta$ → resolution along the incline plane responsible for downward pull on man along the incline plane.

w to find resultant vector ($\vec{R}$):







Using pythagoras theorem

$$|\vec{R}| = \sqrt{(\vec{A}+B\cos\theta)^2+(B\sin\theta)^2}$$

$$|\vec{R}| = \sqrt{A^2 + 2AB\cos\theta + B^2\cos^2\theta + B^2\sin^2\theta}$$

$$|\vec{R}| = \sqrt{A^2 + 2AB\cos\theta + B^2}$$

Direction of $\vec{R}$:

$$\tan\alpha = \frac{B\sin\theta}{\vec{A}+B\cos\theta}$$

$$\Rightarrow \alpha = \tan^{-1}\left(\frac{B\sin\theta}{\vec{A}+B\cos\theta}\right)$$

- if $\theta = 0$ $\rightrightarrows$

$\vec{R} = A+B$, $\alpha = 0$ resultant max

- if $\theta = 180°$ $\leftrightarrows$

$\vec{R} = A-B$, $\alpha = 0$ resultant min

if $\theta = 90°$ $\uparrow\rightarrow$

$\vec{R} = \sqrt{A^2+B^2}$, $\alpha = \frac{B}{A}$

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If $\theta = 120°$ & $|\vec{F_1}| = |\vec{F_2}| = F$  $\rightarrow |\vec{R}| = F$ & $\alpha = 60°$

$$|\vec{R}| = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \times \frac{-1}{2}}$$

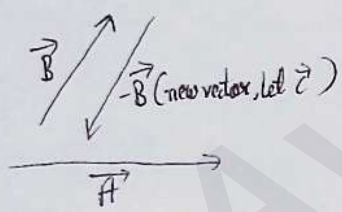
$$\Rightarrow |\vec{R}| = \sqrt{2F_1^2 - F_1^2}$$

$$\Rightarrow |\vec{R}| = \cancel{F_1\sqrt{3}} \quad \sqrt{F_1^2} = F_1 = F$$

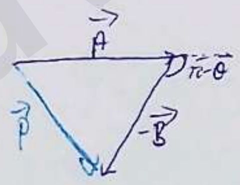
$$\alpha = \frac{F\sin 120}{F + F\cos 120} = \frac{\frac{\sqrt{3}}{2}\cancel{F}}{\cancel{F}(1-\frac{1}{2})} = \frac{\sqrt{3}}{2} \times \frac{2}{1} \overset{\tan^{-1}}{=} (\sqrt{3}) = 60°$$

अगर 2 vector equal in magnitude में है तो उनका resultant exactly बीच में half angle पर होगा लेकिन जरूरी नहीं है magnitude half हो।

Substraction of vectors:



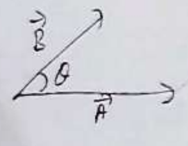
$$\vec{P} = \vec{A} - \vec{B} = \vec{A} + \vec{C}$$

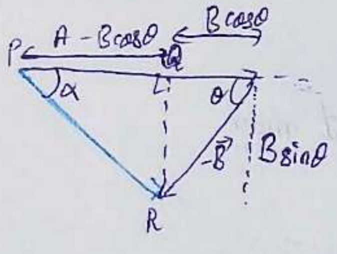


$$\vec{P} = \sqrt{A^2 + B^2 + 2AB\cos(\pi - \theta)}$$

$$\Rightarrow \vec{P} = \sqrt{A^2 + B^2 + (-2AB\cos\theta)}$$

$$\Rightarrow \vec{P} = \sqrt{A^2 + B^2 - 2AB\cos\theta}$$





$$\tan\alpha = \frac{B\sin\theta}{A - B\cos\theta}$$

Substraction में बस $\theta \rightarrow 180 - \theta$ से replace कर देते हैं।

Example :-

(1) ($\xrightarrow[10\,m/s]{v_1}$ east) ($10\,m/s$ $\vec{v_2}$ N of E, 30°)

initial final

find $\Delta V = (\vec{v_2} - \vec{v_1})$

$$|\Delta \vec{V}| = \sqrt{(10)^2 + (10)^2 - \cancel{2}(100) \times \frac{\sqrt{3}}{\cancel{2}}}$$

$$= \sqrt{200 - 100\times\sqrt{3}}$$

$$= 10\sqrt{2-\sqrt{3}}$$

$$= 10 \times \sqrt{2-1.73}$$

$$= 10 \times 0.27 = \boxed{2.7\ m/s.}$$

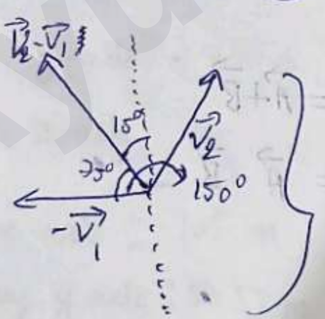
$$\tan\alpha = \frac{10\times\frac{1}{2}}{10 - \frac{10\times\sqrt{3}}{2}} = \frac{5}{10-5\sqrt{3}} = \frac{1}{2-\sqrt{3}} = \cancel{\frac{1}{0.27}}$$

$$= \frac{1}{2} - \frac{1}{\sqrt{3}}$$

$$\alpha = \tan^{-1}\left(\frac{1}{2} - \frac{1}{\sqrt{3}}\right)$$

direction $\tan^{-1}$ के form में है।

$\vec{v_2} + (-\vec{v_1})$ where, $|\vec{v_2}| = |\vec{v_1}|$



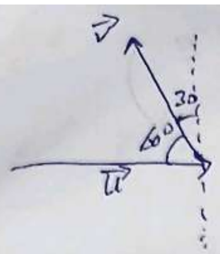
dir^n 15° W of N.

Ans ⇒ $\boxed{2.7\ m/s\ \ 15^\circ\ W\ of\ N}$

(2) $\vec{v_i} = \vec{u}$, $10\,m/s$ along east

$\vec{v} = 10\,m/s$ 30° W of N

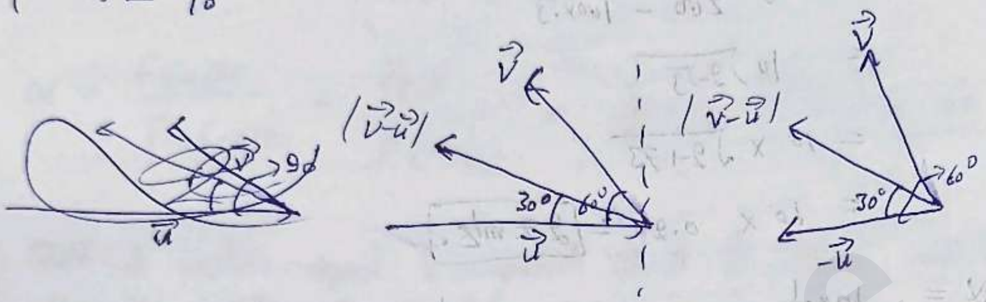
find $\vec{\Delta V} = \vec{v} - \vec{u}$



$$|\vec{v}-\vec{u}| = \sqrt{v^2+u^2-2vu\cos 60}$$

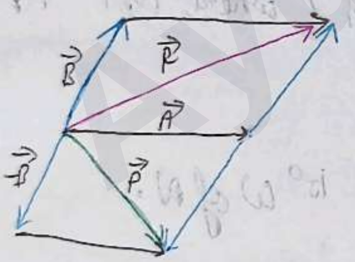
$$= \sqrt{200 - 200\times\tfrac{1}{2}}$$

$$|\vec{v}-\vec{u}| = 10$$



Ans = $\boxed{10\text{ m/s } 30^\circ \text{ N of W}}$

Addition & Substraction in a single diagram:



$$\vec{R} = \vec{A}+\vec{B}$$

$$\vec{P} = \vec{A}-\vec{B}$$

Q) Two vectors $\vec{A}$ and $\vec{B}$:

if $|\vec{A}+\vec{B}| = 2|\vec{A}-\vec{B}|$ & $|\vec{A}| = |\vec{B}|$

Find angle b/w $\vec{A}$ and $\vec{B}$

Sol^n:) $\sqrt{A^2+B^2+2AB\cos\theta} = 2\sqrt{A^2+B^2-2AB\cos\theta}$

$$\Rightarrow A^2+B^2+2AB\cos\theta = 4(A^2+B^2-2AB\cos\theta)$$

$$\Rightarrow \frac{A^2+B^2+2AB\cos\theta}{A^2+B^2-2AB\cos\theta} = \frac{4}{1}$$

Q.P $\Rightarrow \dfrac{2(A^2+B^2)}{4AB\cos\theta} = \dfrac{5}{3}$

$\Rightarrow \dfrac{A^2+B^2}{2AB\cos\theta} = \dfrac{5}{3}$

Let $|\vec{A}| = |\vec{B}| = x$

$\dfrac{2x^2}{2x^2\cos\theta} = \dfrac{5}{3}$

$\Rightarrow 3 = 5\cos\theta$

$\Rightarrow \cos\theta = 5/3$

$\Rightarrow \theta = \cos^{-1}(3/5)$

$\Rightarrow \boxed{\theta = 53^\circ}$

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Cartesian form of vectors :

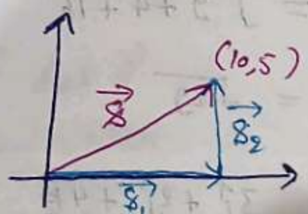
$\hat{i}$ - unit vector along x-axis

$\hat{j}$ - unit vector along y-axis

$\hat{k}$ - unit vector along z-axis

Example :- 10m along x axis : $10\hat{i}$ m

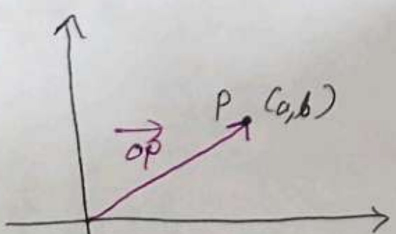
then 5m along y axis : $5\hat{j}$ m



Net displacement $(\vec{s}) = \vec{s_1} + \vec{s_2}$

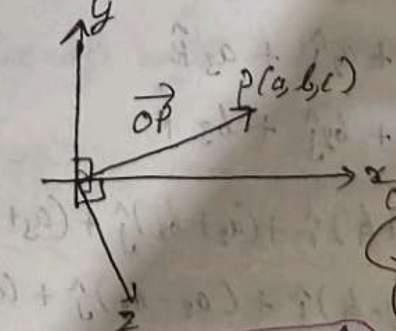
$= (10\hat{i} + 5\hat{j})$ m

#



$\boxed{\vec{OP} = a\hat{i} + b\hat{j}}$

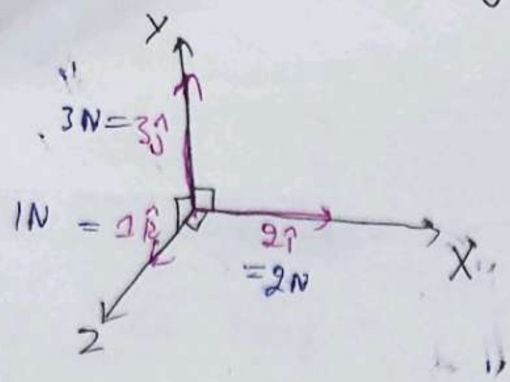
#



x component of OP = a
y ——"—— = b
z ——"—— = c

$\boxed{\vec{OP} = a\hat{i} + b\hat{j} + c\hat{k}}$

Example: force $(\vec{F}) = (2\hat{i} + 3\hat{j} + \hat{k})$ N



Magnitude and direction of Cartesian form of vector:

$$\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

$$\left.|\vec{A}| = \sqrt{(a_1)^2 + (a_2)^2 + (a_3)^2}\ \right\}\text{ magnitude of } \vec{A}$$

$$\vec{A} = |\vec{A}|\ \hat{A}$$

$$\Rightarrow \left.\hat{A} = \frac{\vec{A}}{|\vec{A}|}\ \right\}\text{ unit vector along or dir}^n \text{ of } \vec{A}$$

Example: If force $\vec{F} = 3\hat{i} + 2\hat{j} + 4\hat{k}$, then find magnitude & dir^n of $\vec{F}$.

$$|\vec{F}| = \sqrt{9 + 4 + 16}$$

$$= \sqrt{29}$$

$$\hat{F} = \frac{3\hat{i} + 2\hat{j} + 4\hat{k}}{\sqrt{29}}$$

Addition & Substraction in Cartesian form:

$$\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

$$\vec{B} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$$

$$\vec{A} + \vec{B} = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$$

$$\vec{A} - \vec{B} = (a_1 - b_1)\hat{i} + (a_2 - b_2)\hat{j} + (a_3 - b_3)\hat{k}$$

Example : $\vec{s_1} = 2\hat{i} + \hat{j} - 3\hat{k}$

$\vec{s_2} = \hat{i} + 3\hat{j} + 5\hat{k}$

$\vec{s_3} = 3\hat{i} + \hat{j} - \hat{k}$

find $\vec{s} = \vec{s_1} + \vec{s_2} + \vec{s_3}$

$= 6\hat{i} + 5\hat{j} + \hat{k}$

$|\vec{s}| = \sqrt{36 + 25 + 1}$

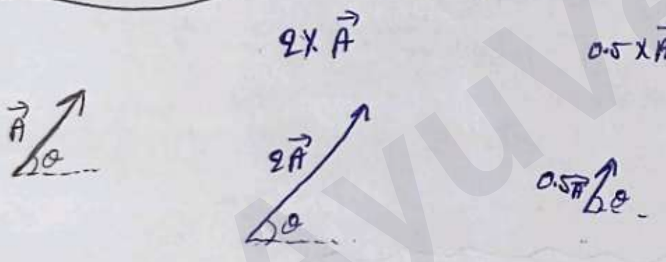
$= \sqrt{62}$

Multiplication of vectors :

① Scalar Multiple
② Dot product / Scalar Product
③ Cross product / vector product

① Scalar multiple (Scaling of vector)

$2 \times \vec{A}$ $\quad 0.5 \times \vec{A}$



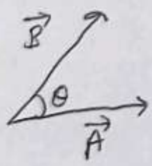
Only magnitude change dirⁿ should be same.

$\vec{s} = k\vec{A}$ ↳ number

$\hat{s} = \hat{A}$

② Scalar Product / Dot Product

$\vec{A} \cdot \vec{B} \equiv \vec{A}$ dot $\vec{B}$

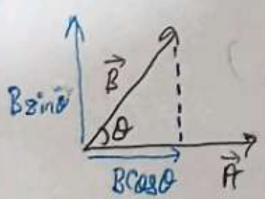


$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta$

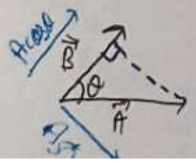
numerical vector

$\cos\theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|}$

The dot product of two vector i.e., $\vec{A}$ & $\vec{B}$ is defined as product of magnitude of both the vectors and cosine of angle b/w them.



$\Longrightarrow A(B\cos\theta) = AB\cos\theta$



$\Longrightarrow (A\cos\theta)B = AB\cos\theta$

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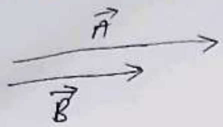
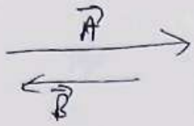
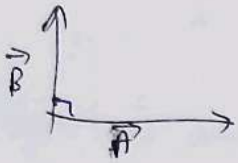


$\vec{A} \wedge \vec{B} = 0^\circ \longrightarrow \vec{A}\cdot\vec{B} = AB\cos 0^\circ = \boxed{AB}$



$\vec{A} \wedge \vec{B} = 180^\circ \longrightarrow \vec{A}\cdot\vec{B} = AB\cos 180^\circ = \boxed{-AB}$



$\vec{A} \wedge \vec{B} = 90^\circ \longrightarrow \vec{A}\cdot\vec{B} = AB\cos 90^\circ = \boxed{0}$

Dot product of a vector with itself $\vec{A}\cdot\vec{A}$ $\rightarrow |\vec{A}|\cdot|\vec{A}|\cos 0^\circ = \boxed{A^2}$

$\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{i}\cdot\hat{k} = \boxed{0}$

$\because \hat{i} \perp \hat{j} \perp \hat{k}$

$\hat{i}\cdot\hat{i} = 1 = \hat{k}\cdot\hat{k} = \hat{j}\cdot\hat{j}$

~~$|\vec{i}|$~~ $\therefore |\hat{i}| = 1.$

Applications of dot product $\Rightarrow$ Calculation of workdone

Dot product in cartesian form:

$\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{B} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{A}\cdot\vec{B} = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k})\cdot(b_1\hat{i} + b_2\hat{j} + b_3\hat{k})$

$= (a_1 b_1\, \hat{i}\cdot\hat{i} + a_1 b_2\, \hat{i}\cdot\hat{j} + a_3 \hat{k}\, b_3 \hat{k}) + (a_2\hat{j}\cdot\hat{k} b \text{---} -) + (- \text{---})$

$= a_1 b_1 + a_3 b_3 + a_2 b_2$

$\vec{A}\cdot\vec{B} = a_1 b_1 + a_2 b_2 + a_3 b_3 = \vec{B}\cdot\vec{A}$

~~Note~~ ~~If $\vec{A} \parallel \vec{B}$; $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{a_1}{c_2}$~~

Note

- If $\vec{A} \parallel \vec{B}$; $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = m$ (constant)
- If $\vec{A} \perp \vec{B}$; $a_1 b_1 + a_2 b_2 + a_3 b_3 = 0$

Example : $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$

$\vec{B} = m\hat{i} + 6\hat{j} + 2\hat{k}$

Find m if (a) $\vec{A} \parallel \vec{B}$
(b) $\vec{A} \perp \vec{B}$

$\vec{a} \parallel \vec{B} \rightarrow \frac{2}{m} = \frac{3}{6} \Rightarrow \boxed{m = 4}$

$\vec{a} \perp \vec{B} \rightarrow 2m + 18 + 2 = 0$

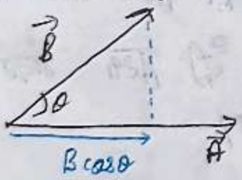
$\Rightarrow 2m = -20$

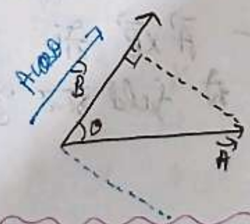
$\Rightarrow \boxed{m = -10}$

- Dot product is distributive also;

$$\vec{A} \cdot (\vec{B} + \vec{C} - \vec{D}) = \vec{A}\cdot\vec{B} + \vec{A}\cdot\vec{C} - \vec{A}\cdot\vec{D}$$

Projection of a vector :





Projection of $\vec{B}$ on $\vec{A} = B\cos\theta$

$= B \frac{\vec{A}\cdot\vec{B}}{|\vec{A}||\vec{B}|}$

$= \vec{B}\cdot\hat{A}$

Projection of $\vec{A}$ on $\vec{B} = A\cos\theta$

$= A \frac{\vec{A}\cdot\vec{B}}{|\vec{A}||\vec{B}|}$

$= \vec{A}\cdot\hat{B}$

Example : $\vec{F} = \hat{i} + 2\hat{j} + \hat{k}$

$\vec{S} = \hat{i} + \hat{j} + \hat{k}$

Find the component of $\vec{F}$ along $\vec{S}$

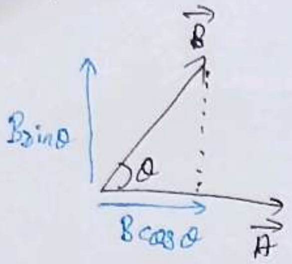
Component of $\vec{F}$ along $\vec{S}$

$\Rightarrow \vec{F}\cdot\hat{S} = (\hat{i} + 2\hat{j} + \hat{k})\left(\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}\right)$

$\Rightarrow \vec{F}\cdot\hat{S} = \frac{1}{\sqrt{3}}(1 + 2 + 1) = \boxed{\frac{4}{\sqrt{3}}}$

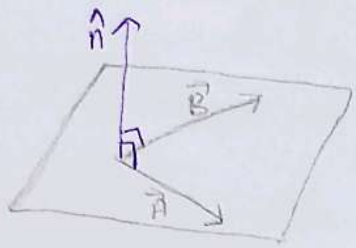
Lec-21

(3) Cross Product or Vector Product



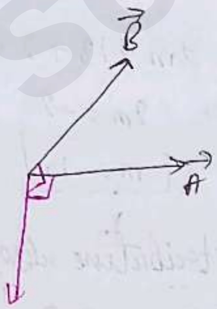
$$\vec{A}\times\vec{B} = |\vec{A}||\vec{B}|\sin\theta\ (\hat{n})$$

where, $\hat{n}$ → unit normal vector
↳ $\perp^r$ to $\vec{A}$ & $\vec{B}$ both.



Direction :- Using right ~~hang~~ hand ~~thumb~~ rule rule :

$$\vec{A}\times\vec{B} \neq \vec{B}\times\vec{A}$$



right hand को उस vector के ओर से दूसरे vector की तरफ move करे
पर dirⁿ आ जाएगा / जैसे - $\vec{A}\times\vec{B}$ में $\vec{A}$ के ओर से $\vec{B}$ की तरफ अंदर की
तरफ फोल्ड करें / Right hand को fold करना है और thumb dirⁿ देगा /

$\vec{A}\times\vec{B} \neq \vec{B}\times\vec{A}$

$|\vec{A}\times\vec{B}| = |\vec{B}\times\vec{A}|$

$\vec{A}\times\vec{B} = -(\vec{B}\times\vec{A})$

The cross product of $\vec{A}$ & $\vec{B}$ is defined as the product of magnitude of both the vector and sin of angle b/w them. & direction of $\vec{A}\times\vec{B}$ is perpendicular to the plane containing $\vec{A}$ and $\vec{B}$ and is according with right hand screw rule

$\vec{A} \parallel \vec{B}$ & $\vec{A} \wedge \vec{B} = 0^\circ \longrightarrow \vec{A} \times \vec{B} = 0$

$\vec{A}$ is antiparallel to $\vec{B}$ & $\vec{A} \wedge \vec{B} = 180^\circ \longrightarrow \vec{A} \times \vec{B} = 0$

$\vec{A} \perp \vec{B}$ & $\vec{A} \wedge \vec{B} = 90^\circ \longrightarrow \vec{A} \times \vec{B} = AB\hat{n} \rightarrow$ Max at 90°

$\vec{A} \times \vec{A} = 0$

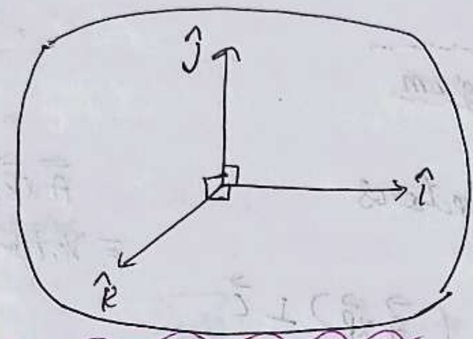
$\hat{i} \times \hat{i} = 0 = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$

$\hat{i} \times \hat{j} = \hat{k}$

$\hat{j} \times \hat{k} = \hat{i}$

$\hat{k} \times \hat{j} = -\hat{i}$

$\hat{k} \times \hat{i} = \hat{j}$



Trick

$\hat{i}\ \hat{j}\ \hat{k}\ \hat{i}\ \hat{j}\ \hat{k}$

$\xrightarrow{+ve}$ / $\xleftarrow{-ve}$ $\Big\}$ $\hat{i} \times \hat{j} = +\hat{k}$, $\hat{k} \times \hat{i} = \hat{j}$

Lec-22

Cross Product in Cartesian form :

$\vec{A} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$

$\vec{B} = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$

$$\begin{array}{ccc} \hat{i} & -\hat{j} & \hat{k} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{array}$$

$$\vec{A} \times \vec{B} = (b_1c_2 - c_1b_2)\hat{i} - (a_1c_2 - a_2c_1)\hat{j} + (a_1b_2 - a_2b_1)\hat{k}$$

Example - $\vec{A} = \hat{i} + 2\hat{j} - \hat{k}$

$\vec{B} = 2\hat{i} + \hat{k}$.

Find $\vec{A} \times \vec{B}$

$$\begin{array}{ccc} \hat{i} & -\hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 2 & 0 & 1 \end{array}$$

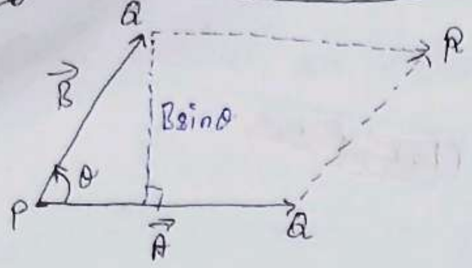
$\vec{A} \times \vec{B} = (2+0)\hat{i} + (1+2)(-\hat{j}) + (0-4)\hat{k}$

$\vec{C} \rightarrow \vec{A} \times \vec{B} = 2\hat{i} - 3\hat{j} - 4\hat{k}$

$\vec{C} \perp \vec{A}$

$\vec{C} \perp \vec{B}$

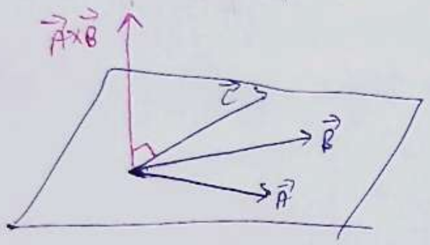
Physical meaning of Cross-Product :



$$\vec{A} \times \vec{B} = AB\cos\theta$$
$$\Rightarrow \vec{A} \times \vec{B} = (\text{Base})(\text{height})$$
$$\Rightarrow \vec{A} \times \vec{B} = \text{Area of parallelogram}$$

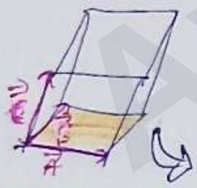
if $\vec{A}, \vec{B}, \vec{C}$ are coplaner vectors

$\vec{A} \times \vec{B} \cdot \vec{C}$ ≡ Vector triplet



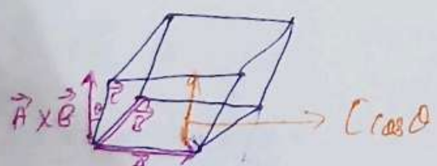
$$(\vec{A} \times \vec{B}) \perp \vec{C}$$
$$\Rightarrow (\vec{A} \times \vec{B}) \cdot \vec{C}$$
$$\Rightarrow \vec{A} \times \vec{B} \cdot \vec{C} = 0$$

If $\vec{A}, \vec{B}, \vec{C}$ are non-coplaner vectors



↳ "Parallelopiped"

$$(\vec{A} \times \vec{B}) \cdot \vec{C} = \text{Volume of parallelopied}$$



Application of Cross Product :
→ Calculation of Torque
↳ Very useful in study of rotation dynamics

Questions

Q-1) If $|\vec{A}\times\vec{B}| = |\vec{A}\cdot\vec{B}|$

Solⁿ: ~~$|\vec{A}\times\vec{B}| \Rightarrow AB\sin\theta = AB\sin$~~

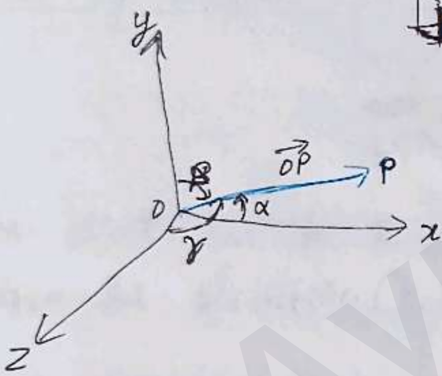
$AB\sin\theta = AB\cos\theta$

$\Rightarrow \tan\theta = 1$

$\Rightarrow \theta = \tan^{-1}(1)$

$\Rightarrow \boxed{\theta = 45^\circ}$

Direction Cosine

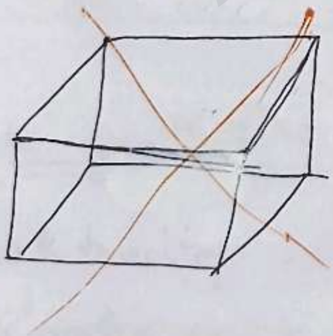


$\vec{OP} \wedge \hat{i} = \alpha$

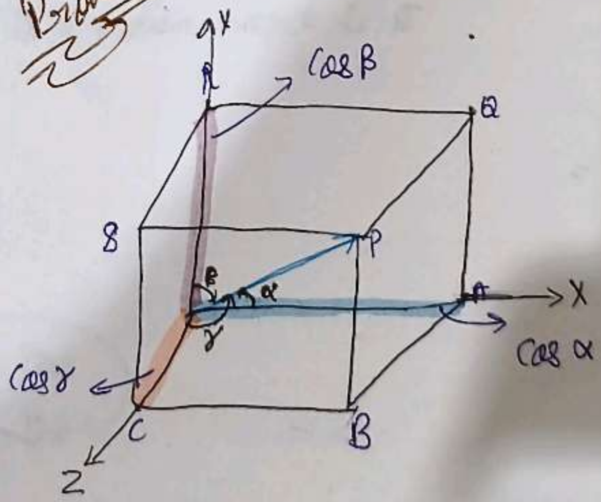
$\vec{OP} \wedge \hat{j} = \beta$

$\vec{OP} \wedge \hat{k} = \gamma$

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$



Prove



$|\vec{OP}| = \sqrt{(x)^2 + (y)^2 + (z)^2}$

$\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma$

$= \frac{OA^2}{OP^2} + \frac{OR^2}{OP^2} + \frac{OC^2}{OP^2}$

$= \frac{x^2 + y^2 + z^2}{(\sqrt{x^2+y^2+z^2})^2} = \boxed{1}$

2. Theorem of change of Vector: